

Beyond Cognitive Load:

**The Projection Principle at Planetary Scale
Manufactured Trust, the Agent Byte, and the Economics of Admission**

The Chatman Equation Thesis Program
(Praxis / Capability Physics)

Genesis — Scale, Paper 04

Abstract

This paper extends the keystone thesis [1] from the theory of a single faithful receipt to the physics of a planetary population of them. We do not restate the keystone’s proofs; we cite them and build forward on three fronts. *First*, we make the Comprehension–Verification Gap quantitative, taking the autonomous build of $\sim 2.5 \times 10^6$ interior tokens as a concrete instance and separating what is *measured* (the interior token count, the receipt field count) from what is *estimated* (per-message hash wall-time), so the gap ratio $\sim 6 \times 10^5$ is stated with its provenance. *Second*, we formalize the industrial practice of *manufactured trust* — the simdjson/QLever treatment — as a general method: a correctness question that Rice’s theorem [2] makes undecidable is retracted onto a decidable *oracle*. We give three oracles and their guarantees: differential agreement as mutual verification (with a false-accept bound), boundary fuzzing of the *total* admission retraction, and mutation-testing of the receipt chain, proving the *Mutant Kill Theorem* — under staged soundness and completeness, a mutant is killed *iff* the validator rejects it at the correct stage. *Third*, we develop the **agent8** projection: an MCP/A2A agent rendered as an eight-bit status byte. We prove that fleet admission is *branchless mask algebra* gating 64 agents per machine word per lane in a fixed seven-instruction reduction, show that a 10^{10} -agent population is exactly 10 GB swept in one memory-bandwidth pass, and give the planetary Fermi estimate (10^8 – 10^{10} admission decisions per second) establishing that only bit-parallel admission is affordable while comprehension-based admission is not. We close by formalizing the *antibody clause*: the overclaim set is empty as an invariant, and the admissible magnitude of any claim is bounded by the mechanism that receipts it — a bound this paper applies reflexively to its own claims. Every abstract object is anchored to running code in the **praxis** repository: the eight-lane **DenialPolarity** word, the eight-bucket **refusal** taxonomy, the staged **PowlReplayVerifier**, the BLAKE3 **OcelCausalFrame** chain, and the **agent8** byte.

Contents

Abstract	i
1 Introduction: From the Gap to the Sweep	1
1.1 What the keystone settled, and what remains	1
1.2 The one-line thesis of this paper	2
2 The Comprehension–Verification Gap, Measured	3
2.1 Measured and estimated quantities	3
2.2 The instance	3
2.3 Boundary cost is invariant under interior growth	4
2.4 Information-Theoretic Scaling and Compression Bounds	4
3 Manufactured Trust as a General Method	7
3.1 The decidable-oracle move	7
3.2 Differential agreement as mutual verification	7
3.3 Fuzzing the admission boundary	8
3.4 Mutation-testing the receipt chain	9
3.5 Categorification of the Denial Monoid and Sheaves	10
3.6 Homological Algebra of Refusal Cascades	12
3.7 Algebraic Consensus and Relativistic Horizons	13
4 Continuous Limits and Planetary Dynamics	15
4.1 The Continuous Limit of the Chatman Equation	15
4.2 Fluid Dynamics of Consequence Conservation	16
4.3 Dynamic System Limits and Feedback Loop Stability	16
5 The Agent Byte and Branchless Fleet Admission	18
5.1 The agent8 byte from the denial lanes	18
5.2 Two layouts and the branchless admission sweep	18
5.3 Footprint and sweep time of a population	20
6 Planetary Capacity and the Economics of Admission	22
6.1 Demand	22
6.2 Supply, two regimes	22
6.3 The true floor is bandwidth, and it is generous	23

7	The Geometry of Swarm Conformance	24
7.1	Infinite-Dimensional Trace Geometry and Farkas' Lemma	24
7.2	Differential Geometry of Swarm Action Fields	25
7.3	Topological Conformance Bounds in Exponential Token Games	25
8	Scale Verification and Reality Addressing	27
8.1	Swarm Scaling and Hierarchical Zero-Defect Bounds	27
8.2	Localized Replay Geometries	28
8.3	Virtual State Transitions and Physical Reality Addressing	28
9	The Antibody Clause, Formalized	30
9.1	Claims, witnesses, and magnitude	30
9.2	Overclaim set and the antibody invariant	30
9.3	Reflexive antibody: this paper's own claims	31
10	Fleet Application and Autonomic Reconciliation	32
10.1	The praxis-retrofit fleet applier	32
10.2	Visual gap measurement and the reconciler residual	33
11	Conclusion	34
A	Notation and Constants	35

Chapter 1

Introduction: From the Gap to the Sweep

1.1 What the keystone settled, and what remains

The keystone thesis [1] settled the *qualitative* shape of trust among agents who cannot comprehend one another. Its results are three. The *Faithful Projection Theorem* [1] establishes that a chained receipt committing law-bearing fields is faithful to lawful consequence over exactly those fields, under collision resistance. The *Comprehension–Verification Gap* [1] establishes that boundary verification of a receipt costs $O(\kappa)$ — independent of interior dimension — while honest replay costs $O(n)$. The *Conservation of Consequence* [1] establishes that every actuated artifact is the image of at least one admitted observation and occupies exactly one receipt-chain position. Together these say: a bounded verifier can trust an unbounded interior by reading a small faithful coordinate of it.

Those theorems are about *one* receipt and *one* verifier. A civilization is not one receipt. It is 10^{10} or 10^{12} agents, each emitting receipts continuously, each obliged to admit or refuse the others’ outputs faster than any of them can be read. Three questions the keystone raised but did not exhaust are the subject of this paper.

- (Q1) **How large is the gap, concretely?** The keystone proved the ratio diverges; it did not pin the constants. We take the 2.5-million-token build as an instance and compute the ratio with a clean line between measured and estimated quantities (Chapter 2).
- (Q2) **Where does the trust actually come from, mechanically?** The keystone named “differential agreement, adversarial fuzzing, invariants, and checked projections” as the engineering practice it was generalizing, but treated them as motivation. We formalize them as a *general method* with provable guarantees (Chapter 3).
- (Q3) **What does admission cost at population scale?** The keystone’s Proposition gave the agent byte and a one-line bandwidth argument. We derive the branchless admission algebra in full, prove its per-word gating, and give the planetary economics that make it the *only* affordable regime (Chapters 5 and 6).

1.2 The one-line thesis of this paper

Everything below is a corollary of a single move, made three times. A question of the form “*is this interior correct / meaningful / trustworthy?*” is, by Rice’s theorem [2], undecidable in general. The industrial answer — and, we argue, the only scalable answer — is never to ask that question. It is to *retract* it onto a decidable surrogate: an equality between two oracles, a total classifier’s labelled output, a hash comparison, a bitwise mask. Each surrogate is cheap, mechanical, and faithful over exactly what it commits. The keystone called this the projection principle for one receipt. This paper shows it is also the mutation oracle, the differential oracle, the fuzzing oracle, and — at population scale — a single AND of eight bit-planes sweeping the planet in one pass over memory.

The gap is why comprehension fails; the sweep is what replaces it.

Structure. Chapter 2 quantifies the gap on the instance and introduces Shannon boundaries. Chapter 3 formalizes manufactured trust, proving the Mutant Kill Theorem and details categories and cohomologies of refusal. Chapter 4 constructs the continuous limit of the Chatman equation. Chapter 5 constructs the agent byte and proves branchless admission. Chapter 6 gives the planetary Fermi estimate. Chapter 7 establishes the scale geometry and separating hyperplanes. Chapter 8 details scaling verification and physical reality addressing. Chapter 9 formalizes the antibody clause. Chapter 10 details retrofit apps and reconcilers. Chapter 11 concludes. Throughout, results proved are theorems; quantities that must be estimated are set as *Estimates* and labelled as such, following the keystone’s discipline of claiming only what is true.

Chapter 2

The Comprehension–Verification Gap, Measured

The keystone proved that the ratio of comprehension cost to boundary-verification cost diverges as the interior grows. Divergence is a statement about a limit; a civilization runs on constants. This chapter supplies the constants for the canonical instance and, in doing so, fixes a methodological rule for the whole series: *every reported number is tagged measured or estimated*.

2.1 Measured and estimated quantities

Definition 2.1 (Instance quantities). For an executed manufacture σ we distinguish:

- $T(\sigma)$, the *interior token count* — the number of machine-generated tokens in the full trace. This is *measured*: it is a count of logged records.
- κ , the *boundary field count* — the number of independently manipulable chunks in the receipt projection $\pi(\sigma)$. This is a *design constant*, fixed by the frame schema, not by the run.
- t_H , the *per-frame recomputation wall-time* — the time to hash one frame and compare to the claimed chain value. This is *estimated* from published throughput of the hash primitive; it is not claimed as measured here.

Remark. The tag matters because the doctrine’s credibility is exactly its refusal to launder estimates as measurements. T and κ are hard numbers; t_H is a Fermi input. We never add a measured and an estimated number without saying so.

2.2 The instance

The canonical instance is the autonomous build of the keystone’s Chapter 6: a fleet performed exploration, planning, and implementation whose interior consumed $T(\sigma) \approx 2.5 \times 10^6$ tokens (measured), no meaningful fraction of which any agent or human read. Its receipt projection has the schema:

$$\pi(\sigma) = \left(\underbrace{\text{verdict}}_{1 \text{ bit}}, \underbrace{h_+}_{256 \text{ bits}}, \underbrace{\varphi}_{1 \text{ scalar}}, \underbrace{r}_{1 \text{ reason word}} \right), \quad \kappa = 4 \text{ fields (design)}.$$

The verdict is one bit; h_+ is the 256-bit BLAKE3 chain head of the `OcelCausalFrame` sequence; $\varphi \in [0, 1]$ is the `PowlReplayVerifier` fitness; r is one of the eight refusal categories (`Identity`, `Capacity`, `Topology`, `Temporal`, `Lifecycle`, `Authorization`, `Prerequisites`, `Reserved`) or empty on accept.

Theorem 2.1 (Instance gap). *Let comprehension cost be $\text{Comp}(\sigma) = \Theta(T(\sigma))$ and boundary-verification cost be $\text{Ver}_\partial(\sigma) = \Theta(\kappa)$. For the instance,*

$$\frac{\text{Comp}(\sigma)}{\text{Ver}_\partial(\sigma)} = \Theta\left(\frac{T(\sigma)}{\kappa}\right) = \Theta\left(\frac{2.5 \times 10^6}{4}\right) \approx 6.25 \times 10^5.$$

Both inputs to the ratio are measured/design constants; the ratio is therefore a measured consequence, not an estimate.

Proof. Immediate from the gap ratio definitions with the two constants of Definition 2.1 substituted. Comprehension must ingest $\Theta(T)$ tokens; boundary verification reads the $\kappa = 4$ receipt fields and applies the fixed verifier V . The quotient is T/κ ; substitution gives the figure. No estimated quantity enters. $\square$

Remark. Six hundred thousand is not the ratio of trace bytes to receipt bytes; it is the ratio of what a verifier would have to *comprehend* to the number of independent chunks it actually *reads*. The full trace can still be replayed at honest cost $O(n)$; the gap concerns comprehension versus boundary inspection, and it grows without bound as T grows while κ stays fixed.

2.3 Boundary cost is invariant under interior growth

Proposition 2.2 (Interior-independence of the boundary). *Let $\{\sigma_i\}$ be manufactures sharing the frame schema of Definition 2.1 but with interior token counts $T(\sigma_i) \rightarrow \infty$. Then $\text{Ver}_\partial(\sigma_i) = \Theta(\kappa)$ is constant in i , and the estimated per-message wall-time $\kappa \cdot (t_{\text{cmp}}) + c \cdot t_{\text{H}}$ (for a constant number c of recomputed frames in a spot-check) does not depend on $T(\sigma_i)$.*

Proof. The projection π has codomain of size $\leq \kappa$; the verifier V reads only $\pi(\sigma_i)$. Neither the field count nor V 's input size is a function of T . A spot-check recomputes a fixed number c of frames, each $O(1)$, so its cost is $c \cdot t_{\text{H}}$, again independent of T . $\square$

Estimate 2.1 (Per-message wall-time). With BLAKE3 throughput on the order of 1–3 GB/s per core and a frame of a few hundred bytes, $t_{\text{H}} \sim 10^{-7}$ s per frame; a four-field boundary compare is a handful of machine words, $\ll 10^{-7}$ s. A per-message verification that inspects the boundary and recomputes $c \sim 10$ spot frames costs $\sim 10^{-6}$ s. These are order-of-magnitude figures from published primitive throughput, labelled as estimates; the point they support — constancy in T — is the proven Proposition 2.2, not the estimate.

2.4 Information-Theoretic Scaling and Compression Bounds

To scale the gap to trillion-agent swarms ($N = 10^{12}$), we model global state transition sparsity and compression limits of hierarchical commitments. Let $\mathcal{S}$ be the finite set of local agent states, and $S_t = (s_{1,t}, \dots, s_{N,t}) \in \mathcal{S}^N$ represent the global swarm state. The transition sparsity is $k_t = |\{i \in [1, N] \mid s_{i,t} \neq s_{i,t-1}\}| \ll N$. The global state commitment R_t is generated via a

hierarchical Merkle Mountain Range (MMR) [29] where the leaf nodes are $v_{i,0}^{(t)} = H(i \| s_{i,t})$ and parent nodes are recursively defined. Each state transition is committed to a rolling hash chain $h_t = H(h_{t-1} \| \text{fr}_t)$ with $\text{fr}_t = \langle \theta_t, R_t \rangle$, where θ_t is low-entropy metadata.

Theorem 2.3 (Swarm Receipt Compression Bounds). *Let $N = 10^{12}$ and transition sparsity $k_t \ll N$. The information-theoretic compression ratio C_t of the receipt chain relative to the execution trace is:*

$$C_t = \frac{H(\Sigma_t)}{H(\mathcal{R}_t)} = \frac{H(S_0) + \sum_{j=1}^t H(S_j | S_{j-1})}{256 \cdot t + H(\theta_{1..t})}.$$

Under sparse transitions, the conditional entropy is bounded by $H(S_t | S_{t-1}) \leq \log_2 \binom{N}{k_t} + k_t H(S)$. Applying Stirling's approximation, the compression ratio scales as:

$$C_t \approx \Omega \left(\frac{N \log |\mathcal{S}|}{256 + H(\theta_t)} \right)$$

which diverges to infinity as $N \rightarrow \infty$.

Proof. The mapping $\pi : \Sigma_t \rightarrow h_t$ is a deterministic projection. By the data processing inequality, $H(h_t) \leq H(\Sigma_t)$. The collision resistance of H ensures that any trace perturbation $\tilde{\Sigma}_t \neq \Sigma_t$ yields a different hash chain $h_t \neq \tilde{h}_t$, except with negligible probability $\varepsilon(\lambda) \leq 2^{-256}$. Thus, h_t is a mathematically binding representative of Σ_t of size 256 bits, independent of N . The compression ratio is $\Omega(N)$ for any fixed time step. $\square$

Theorem 2.4 (Comprehension-Verification Gap at Scale). *Let $C_C(\sigma)$ be the cognitive cost of comprehending the trace $\sigma \in \Sigma$, and $C_V(r)$ be the cognitive cost of verifying the receipt $r \in \mathcal{R}$. If $C_C(\sigma) = \Omega(N \cdot T)$ and $C_V(r) = O(\kappa)$ for context capacity limit $\kappa \approx 4$, then the comprehension-verification gap Γ_N satisfies:*

$$\Gamma_N = \frac{C_C(\sigma)}{C_V(r)} = \Omega \left(\frac{N \cdot T}{\kappa} \right).$$

At trillion-agent scale ($N = 10^{12}$), $\Gamma_N \rightarrow \infty$, rendering verification cognitive-load eligible under κ while guaranteeing that any safety violation in the trace σ is detected.

Proof. Let σ be an invalid trace containing a safety violation. The local admission map α evaluates obligations g_i . If a violation occurs, the total denial word $d(\sigma) \neq \mathbf{0}$. If $d(\sigma) \neq \mathbf{0}$, the conformance fitness $\varphi(\sigma) < 1$. The human operator checks: (1) Is verdict == pass? (2) Is $\varphi(\sigma) == 1.0$? (3) Is $d(\sigma) == \mathbf{0}$? (4) Is h_T signed by the cryptographic key k^* ? The number of checks is $4 \leq \kappa$. If any check fails, the operator rejects; if all pass, the operator accepts. By the collision resistance of the hash chain and signature security, the probability that the operator accepts an invalid trace is bounded by $\varepsilon(\lambda)$. Thus, verification is achieved in $O(\kappa)$ cognitive steps, proving that a single human operator can verify a trillion-agent action trace. $\square$

Theorem 2.5 (Cryptographic Admission Safety Bound). *If the obligation battery $G = \{g_1, \dots, g_m\}$ contains cryptographic signature checks with security parameter $\lambda = 256$, then for any random noise distribution $\eta \neq 0$ such that $\tilde{o} = o + \eta$, the false admission probability p_{fa} is bounded by:*

$$p_{\text{fa}} \leq 2^{-\lambda}$$

Furthermore, the false refusal probability p_{fr} is:

$$p_{\text{fr}} = 1 - \Pr[\eta = 0]$$

Proof. Let o be a valid observation with a valid signature. Any noise perturbation $\eta \neq 0$ results in a modified observation $\tilde{o} = o + \eta$. Because the signature check is cryptographic and modeled as a random oracle or cryptographically secure function, the probability that the signature of o is also valid for $\tilde{o}$ is at most the probability of finding a collision or forging a signature under parameter λ , which is $2^{-\lambda}$. Thus, the false admission probability $p_{\text{fa}} = \Pr[G(\tilde{o}) = 1 \mid \eta \neq 0] \leq 2^{-\lambda}$. A false refusal occurs if a valid observation is perturbed, so any non-zero noise $\eta \neq 0$ causes the signature check to fail. The probability of this event is exactly the probability that the noise is non-zero, i.e., $p_{\text{fr}} = 1 - \Pr[\eta = 0]$. $\square$

Theorem 2.6 (Chatman Shannon Partition Bound). *Let C_{jk} be the channel capacity between partition j and partition k . Let $H(L_j \mid L_{j,\text{base}})$ be the entropy generation rate of the local process log in partition j . Reconciliation via a three-way merge is possible without loss of consistency if and only if the mutual information rate of the communication channel satisfies:*

$$I(O_j; O_k) \geq \max(H(L_j \mid L_{j,\text{base}}), H(L_k \mid L_{k,\text{base}})) - H(\Delta_{\text{resolvable}})$$

If the channel capacity falls below the coherence threshold:

$$\min(C_{jk}) < \mathcal{H}_{\text{drift}}$$

*then the partitions drift, and the system must fire the refusal lane **Topology** or **Temporal** in the denial word $d(o) \neq \mathbf{0}$, forcing the local admission retraction to refuse $\alpha(o) = \perp \implies \mu(\alpha(o)) = \perp$, halting local manufacture.*

Proof. Let partitions j and k generate local logs L_j and L_k independently. By Shannon's source coding theorem [8], the minimum rate required to transmit the updates of log L_j relative to its base $L_{j,\text{base}}$ is its conditional entropy rate $H(L_j \mid L_{j,\text{base}})$. To reconcile the partitions without losing consistency, the channel must transmit all non-resolvable changes. The information-theoretic limit on the rate of resolvable change is the mutual information rate $I(O_j; O_k)$ augmented by the resolvable difference entropy $H(\Delta_{\text{resolvable}})$. Hence, consistency is preserved if and only if the channel mutual information rate satisfies the lower bound. If the channel capacity $\min(C_{jk}) < \mathcal{H}_{\text{drift}}$, the rate of information transfer is strictly less than the drift rate of the states. In this case, the states drift beyond the coherence threshold, which triggers a **Topology** or **Temporal** refusal in the denial word $d(o) \neq \mathbf{0}$. By the structure of the admission retraction, a non-zero denial word forces the admission mapping to return $\perp$. The manufacturing morphism μ maps $\perp$ to $\perp$, halting local manufacture and preventing safety divergence. $\square$

Chapter 3

Manufactured Trust as a General Method

Engineers already trust software they cannot comprehend. A branchless vector parser classifies sixty-four input bytes per instruction [3]; an indexed triple store answers a query over 10^{11} facts by a join plan no one reconstructs [4]. Neither is trusted by reading it. Both are trusted because a battery of *decidable oracles* stands in for the undecidable question of correctness. This chapter turns that practice into theory. The unifying claim is the keystone’s, sharpened: each oracle is an *admission retraction* — a total, terminating, decidable surrogate for a semantic question Rice’s theorem forbids us to decide directly.

3.1 The decidable-oracle move

Definition 3.1 (Correctness question and decidable oracle). Let $f : D \rightarrow R$ be an implementation and $S \subseteq D \times R$ a specification. The *correctness question* is $Q_S(f) \equiv \forall x (x, f(x)) \in S$. A *decidable oracle* for f on a finite test set $X \subseteq D$ is a total computable predicate $\Omega : X \rightarrow \{0, 1\}$ such that $\Omega(x) = 1$ is intended to witness $(x, f(x)) \in S$, and Ω terminates on every $x \in X$.

Proposition 3.1 (Why an oracle is necessary, not a convenience). *For S a non-trivial semantic property, $Q_S(f)$ is undecidable. Hence no procedure decides $Q_S(f)$ for arbitrary f ; any mechanical trust in f must be trust in some Ω evaluated on a finite X , together with an argument bounding the residual risk on $D \setminus X$.*

Proof. $Q_S(f)$ quantifies a non-trivial semantic predicate over the language computed by f ; by Rice’s theorem [2] it is undecidable. A total Ω on finite X always terminates and is therefore not a decision procedure for Q_S ; it is a retraction of Q_S onto the decidable fragment “ f passes Ω on X .” Trust in f factors through Ω and X . $\square$

Proposition 3.1 is the whole chapter in one line: the three methods below are three constructions of Ω , each with a different residual-risk argument.

3.2 Differential agreement as mutual verification

The simdjson/QLever pattern’s first oracle is *differential*: run a second, independent implementation and compare outputs. Agreement is decidable; correctness is not.

Definition 3.2 (Differential oracle). Let $f, g : D \rightarrow R$ target the same S . The *differential oracle* is $\Omega_{f,g}(x) = [f(x) = g(x)]$. A test *passes* at x iff $\Omega_{f,g}(x) = 1$.

A pass can be wrong in only one way: both implementations wrong *and* wrong identically. The next proposition bounds that event under an explicit independence assumption.

Proposition 3.2 (False-accept bound for differential agreement). *Fix $x \in D$. Suppose f, g err at x with probabilities p_f, p_g , that their errors are independent, and that conditioned on both erring their wrong outputs are drawn from a set of at least $M \geq 2$ distinguishable values with collision probability $\leq 1/M$. Then the probability that x passes the differential oracle yet $(x, f(x)) \notin S$ satisfies:*

$$\Pr[\Omega_{f,g}(x) = 1 \wedge (x, f(x)) \notin S] \leq \frac{p_f p_g}{M}.$$

Over an independent test corpus X of size n sampling the same regime, the probability that some silent-agreement error survives all n tests is at most $n p_f p_g / M$, and the probability that a fixed silently-wrong input is missed by a corpus that would have exercised it with per-input hit rate q is $(1 - q)^n \rightarrow 0$.

Proof. A false accept at x requires $f(x) \neq S$ -correct value *and* $g(x) = f(x)$. The first has probability p_f . Given both err (probability $p_f p_g$ by independence), agreement requires the two wrong outputs to coincide, probability $\leq 1/M$. If f is correct at x but g wrong, agreement fails and no false accept occurs; if f wrong but g correct, agreement fails likewise. Hence the joint event has probability $\leq p_f p_g / M$. The union bound over n corpus points gives $n p_f p_g / M$. The miss bound is the probability a Bernoulli- q trigger never fires in n draws, $(1 - q)^n$. $\square$

Remark. $\Omega_{f,g}$ admits f 's output on the evidence of g and simultaneously g 's on the evidence of f : neither comprehends the other's interior, exactly as two agents at fleet scale do not. The independence assumption is the load-bearing one — shared bugs (same author, same library, same wrong spec reading) violate it, which is why the practice pairs implementations of *maximally different provenance* (a scalar reference against a SIMD kernel; a naive triple scan against an indexed join). We state the assumption rather than hide it; the antibody clause (Chapter 9) forbids claiming a bound whose hypothesis is unmet.

3.3 Fuzzing the admission boundary

The second oracle probes not μ but α itself: it searches for inputs where the admission retraction misbehaves. Here the projection principle pays a structural dividend. Because α is *total* — its totality is mechanized by wildcard-free `match` on a closed refusal enum (the foundations paper's Proposition on mechanized totality, realized in `refusal.rs` where `RefusalScenario::category` and `denial_lane` cover every variant with no catch-all arm) — every input has a defined, labelled outcome.

Definition 3.3 (Admission-boundary fuzzing oracle). Let $\alpha : \mathcal{O} \rightarrow \mathcal{O}^* \cup \{\perp\}$ be the total admission retraction. The *fuzzing oracle* Ω_∂ maps a generated input $o \in \mathcal{O}$ to 1 iff $\alpha(o)$ *terminates* and yields either an admitted $x \in \mathcal{O}^*$ with a well-formed artifact $\mu(x)$ or a refusal $\perp$ carrying a category in the eight-bucket taxonomy; $\Omega_\partial(o) = 0$ iff α crashes, diverges, or returns an unlabelled outcome.

Proposition 3.3 (Fuzzing soundness for a total retraction). *If α is total and terminating with codomain $\mathcal{O}^* \cup \{\perp\}$ and $\perp$ is refusal-with-category (refusal-as-data), then for every $o \in \mathcal{O}$, $\Omega_{\partial}(o) = 1$, and any observed $\Omega_{\partial}(o) = 0$ is a genuine defect in the retraction, not a property of the input.*

Proof. Totality gives termination on all o ; the closed codomain gives, for each o , either $x \in \mathcal{O}^*$ (whence $\mu(x)$ is defined by boundedness) or $\perp$ with a category, since every refusal path maps through the wildcard-free classifier. Thus the disjunction defining $\Omega_{\partial}(o) = 1$ holds for every o , so $\Omega_{\partial} \equiv 1$ on a correct implementation. Contrapositively, an observed 0 — a crash, a hang, or an unlabelled return — contradicts totality or closedness and hence witnesses an implementation defect. $\square$

Remark. Fuzzing cannot and does not test “did the agent understand the observation” — that is undecidable (Proposition 3.1). It tests that the *retraction* is total, terminating, and labelling: exactly the properties the keystone requires of α . This is why fuzzing an admission boundary is cheap and conclusive where fuzzing for “meaning” would be neither. The boundary between admitted and refused is a decidable surface; the fuzzer walks it.

3.4 Mutation-testing the receipt chain

The third oracle turns the tools on the verifier itself. *Mutation testing* [5, 6] injects a small fault (a *mutant*) and asks whether the test suite — here, the staged **PowlReplayVerifier** — detects it. A mutant that no test detects is *surviving*; one that is detected is *killed*. We formalize the verifier as a pipeline of stages and prove that killing is exactly rejection at the mutant’s proper stage.

The **praxis** verifier runs four stages in order: *recompute* (rehash each frame), *linkage* (each frame’s back-pointer equals the predecessor’s chain head), *monotonic* (sequence indices strictly increase), and *POWL token replay* (the marking geometry, fitness $\varphi = 1$ iff no disabled firing). Model this abstractly.

Definition 3.4 (Staged validator). A *staged validator* is a pipeline $V = (V_1, \dots, V_k)$ where each stage $V_i : \mathcal{S} \rightarrow \{\text{pass}\} \cup \{\text{reject}_i\}$ tests a decidable invariant $I_i \subseteq \mathcal{S}$: $V_i(s) = \text{pass} \iff s \in I_i$. The pipeline accepts s iff every stage passes, i.e. $s \in \bigcap_i I_i$; on rejection it reports the least i with $s \notin I_i$. Stage V_i is *sound* if $V_i(s) = \text{reject}_i \Rightarrow s \notin I_i$ and *complete* if $s \notin I_i \Rightarrow V_i(s) = \text{reject}_i$.

Definition 3.5 (Mutation operator and correct stage). A *mutation operator* m maps a valid subject $s^* \in \bigcap_i I_i$ to a mutant $m(s^*)$ that violates a non-empty set of invariants. The *correct stage* of m is

$$s(m) = \min\{i : m(s^*) \notin I_i\},$$

the first invariant the mutant breaks. A mutant is *killed* by V , written $\text{Kill}(m) = 1$, iff V rejects $m(s^*)$.

Theorem 3.4 (Mutant Kill Theorem). *Let $V = (V_1, \dots, V_k)$ be a staged validator in which every stage is sound and complete (Definition 3.4), and let m be a mutation operator with correct stage $s(m)$ (Definition 3.5). Then*

$$\text{Kill}(m) = 1 \iff V_{s(m)} \text{ rejects } m(s^*),$$

and moreover the least rejecting stage reported by V equals $s(m)$. Equivalently: a mutant is killed if and only if the validator rejects it at the correct stage.

Proof. ($\Leftarrow$) If $V_{s(m)}$ rejects $m(s^*)$ then V rejects (some stage rejected), so $\text{Kill}(m) = 1$.

($\Rightarrow$) Suppose $\text{Kill}(m) = 1$, so some stage rejects $m(s^*)$; let j be the least such. By soundness of V_j , $m(s^*) \notin I_j$. By completeness of every stage $V_1, \dots, V_{j-1}$, none of which rejected, $m(s^*) \in I_i$ for all $i < j$. Hence j is the least index with $m(s^*) \notin I_i$, which is by definition $s(m)$. Thus $j = s(m)$ and the rejecting stage is $V_{s(m)}$.

For the “moreover”: V reports the least rejecting index, which the argument identifies as $s(m)$. Finally, killing occurs at all iff $s(m)$ exists, i.e. iff m violates some invariant; a mutation that violates no I_i leaves $m(s^*) \in \bigcap_i I_i$, is accepted, and is (correctly) not killed — it is an *equivalent mutant*, semantically identical modulo the committed invariants. $\square$

Corollary 3.5 (Diagnostic completeness of the chain). *Under Theorem 3.4, mutation testing of a sound-and-complete staged validator is not merely a pass/fail gate: the index of the killing stage certifies which law-bearing invariant the mutation broke. A mutant that flips a chain byte is killed at recompute; one that rewires a back-pointer at linkage; one that reorders frames at monotonic; one that forces a disabled transition at POWL replay. Surviving mutants are exactly the equivalent mutants (no committed invariant broken), so a non-empty surviving set that is not equivalent is a proof that the frame under-commits — the Faithful Projection converse of [1, Thm. 3.2].*

Proof. By Theorem 3.4, the stage index j at which the validator V rejects a mutant $m(s^*)$ is exactly the correct stage $s(m) = \min\{i : m(s^*) \notin I_i\}$. Because the stages V_1, V_2, V_3, V_4 correspond respectively to the disjoint structural invariants *recompute*, *linkage*, *monotonic*, and *POWL replay*, the reported rejecting index j uniquely identifies the specific law-bearing invariant I_j violated.

For the converse, if a non-equivalent mutant $m(s^*)$ survives ($\text{Kill}(m) = 0$), then $m(s^*) \in \bigcap_i I_i$. Since the mutant is semantically distinct from s^* yet accepted by all stages, the committed invariants fail to capture the semantic difference, proving that the frame under-commits. This is the contrapositive of the Faithful Projection property ([1, Thm. 3.2]), which guarantees that distinct semantic outcomes project to distinct, rejectable receipt fields. $\square$

Remark. Differential agreement (Def. 3.2), boundary fuzzing (Def. 3.3), and mutation kill (Def. 3.5) are three instances of the decidable-oracle move (Def. 3.1): each replaces an undecidable “is it correct” with a decidable check — an equality, a totality witness, a staged rejection. This is the projection principle of [1] applied to the *trust process* rather than the *trust artifact*. The receipt projects the interior; these oracles project the correctness question.

3.5 Categorification of the Denial Monoid and Sheaves

At the scale of 10^{12} agents, the dimension n of the denial monoid $D = (\{0, 1\}^n, \vee, \mathbf{0})$ grows without bound, rendering bit-vector representations computationally and communicatively intractable. Categorification replaces elements $d \in D$ with objects in a symmetric monoidal category **Den**.

Definition 3.6 (Denial Category). A *Denial Category* **Den** is a skeletal symmetric monoidal category $(\mathbf{Den}, \otimes, I)$ where:

1. The objects are refusal witness bundles.
2. The monoidal product $\otimes$ is idempotent, meaning there exists a natural isomorphism $\delta_A : A \rightarrow A \otimes A$ for every object $A \in \mathbf{Den}$, satisfying symmetric monoidal coherence.
3. For any two objects $A, B \in \mathbf{Den}$, there exists at most one morphism $f : A \rightarrow B$, which exists if and only if $A \preceq B$ (refusal refinement).
4. The monoidal unit I represents the clean admission state $\mathbf{0}$, satisfying $A \otimes I \cong A$.

Theorem 3.6 (Categorical Idempotence of Denial). *Let S be a set of refusal lanes, and let $\mathbf{Den}$ be the category whose objects are subsets of S with finite support, where a morphism $f : A \rightarrow B$ exists if and only if $A \subseteq B$. Then $(\mathbf{Den}, \cup, \emptyset)$ is a symmetric monoidal category that is idempotent up to natural isomorphism.*

Proof. Let A, B, C be subsets of S . The monoidal product is defined by the set union, $A \otimes B = A \cup B$, and the monoidal unit is the empty set, $I = \emptyset$. Since set union is associative and has $\emptyset$ as a identity element, the associativity isomorphism and unit isomorphisms are identity morphisms. They satisfy the Mac Lane pentagon [10] and triangle equations because the hom-sets contain at most one morphism. The symmetry isomorphism $s_{A,B} : A \cup B \rightarrow B \cup A$ is the identity morphism, satisfying the symmetry hexagon equations. For any object $A \in \mathbf{Den}$, we have $A \cup A = A$. The diagonal map $\delta_A : A \rightarrow A \cup A$ is the identity morphism. Because δ_A is an identity, it is a natural isomorphism. Coherence diagrams commute directly because all arrows are identity morphisms. $\square$

To model a distributed trillion-agent swarm, we represent the spatial distribution of failure lanes using a topological space X of refusal lanes, where open sets represent causally correlated failure domains.

Theorem 3.7 (Sheaf of Swarm Denials). *Let X be the topological space of refusal lanes. The assignment $\mathcal{D} : U \mapsto \mathbf{Den}_U$ of the local denial category on the open set $U \subseteq X$ defines a sheaf of symmetric monoidal categories on X , whose category of global sections $\Gamma(X, \mathcal{D})$ characterizes the global refusal state of the swarm.*

Proof. We show $\mathcal{D}$ satisfies the gluing axioms for objects and morphisms. (1) Morphism Monopole: Let $U \subseteq X$ be open, and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of U . Let $A, B \in \mathcal{D}(U)$. If there are morphisms $f, g : A \rightarrow B$ such that their restrictions $f|_{U_i} = g|_{U_i}$ for all $i \in I$, then $f = g$. Since $\mathcal{D}(U)$ is a poset category, $|\text{Hom}(A, B)| \leq 1$, so if a morphism exists, it is unique and f equals g . (2) Gluing of Morphisms: If we have local morphisms $f_i : A|_{U_i} \rightarrow B|_{U_i}$ such that they agree on overlaps, they must glue to a unique global morphism $f : A \rightarrow B$. Agreeing on overlaps means $A|_{U_i \cap U_j} \subseteq B|_{U_i \cap U_j}$ for all i, j . Since A and B are sheaves of sets on X , the containment relation $A(U_i) \subseteq B(U_i)$ for all i implies $A(U) \subseteq B(U)$ by the gluing property of subset sheaves, so $f : A \rightarrow B$ exists globally. (3) Gluing of Objects: Let $\{A_i \in \mathcal{D}(U_i)\}_{i \in I}$ be a family of local refusal objects such that for all $i, j \in I$, $A_i|_{U_i \cap U_j} \cong A_j|_{U_i \cap U_j}$. Since the categories are posets, isomorphism implies equality, $A_i|_{U_i \cap U_j} = A_j|_{U_i \cap U_j}$. By the sheaf property of the underlying sets, there exists a unique global object $A \in \mathcal{D}(U)$ such that $A|_{U_i} = A_i$ for all i . $\square$

3.6 Homological Algebra of Refusal Cascades

In a swarm network of 10^{12} agents, a refusal is not isolated: it propagates along communication links, causing cascade refutations. Let the swarm topology be modeled as a simplicial complex $\mathcal{K}$, where vertices $\mathcal{K}_0$ represent individual agents, and p -simplices $\mathcal{K}_p$ represent cliques of $p + 1$ communicating agents.

Definition 3.7 (Refusal Cochain Complex). Let R be an abelian group of refusal values. The space of p -cochains $\mathcal{R}^p = \text{Map}(\mathcal{K}_p, R)$ represents the refusal configurations on p -simplices. The refusal differential $d_p : \mathcal{R}^p \rightarrow \mathcal{R}^{p+1}$ is defined by:

$$(d_p \omega)(v_0, \dots, v_{p+1}) = \sum_{j=0}^{p+1} (-1)^j \omega(v_0, \dots, \widehat{v}_j, \dots, v_{p+1})$$

where $\omega \in \mathcal{R}^p$, and $\widehat{v}_j$ denotes the omission of the vertex v_j .

Theorem 3.8 (Refusal Differential Nilpotence). *The sequence of refusal cochains forms a cochain complex, i.e., the composite differential satisfies:*

$$d_{p+1} \circ d_p = 0.$$

Proof. Let $\omega \in \mathcal{R}^p$. Evaluating the composition $(d_{p+1} d_p \omega)$ on a $(p + 2)$ -simplex $(v_0, \dots, v_{p+2})$ gives:

$$\begin{aligned} (d_{p+1} d_p \omega)(v_0, \dots, v_{p+2}) &= \sum_{i=0}^{p+2} (-1)^i (d_p \omega)(v_0, \dots, \widehat{v}_i, \dots, v_{p+2}) \\ &= \sum_{i=0}^{p+2} (-1)^i \left(\sum_{j < i} (-1)^j \omega(v_0, \dots, \widehat{v}_j, \dots, \widehat{v}_i, \dots, v_{p+2}) \right. \\ &\quad \left. + \sum_{j > i} (-1)^{j-1} \omega(v_0, \dots, \widehat{v}_i, \dots, \widehat{v}_j, \dots, v_{p+2}) \right). \end{aligned}$$

In this double summation, every term of the form $\omega(v_0, \dots, \widehat{v}_i, \dots, \widehat{v}_j, \dots, v_{p+2})$ with $i < j$ appears twice. Once from the first term when the outer index is j and the inner index is i (with coefficient $(-1)^{i+j}$). Once from the second term when the outer index is i and the inner index is j (with coefficient $-(-1)^{i+j}$). Summing these coefficients yields zero, proving that $d_{p+1} \circ d_p = 0$. $\square$

The nilpotence of the differential ensures that refusal boundaries are stable. The refusal cohomology groups are $H^p(\mathcal{R}^*) = \ker d_p / \text{im } d_{p-1}$.

Theorem 3.9 (Refusal Cascade Cohomology). *Let $\mathcal{K}$ be the agent network, and $\mathcal{R}^*$ be the cochain complex of refusal states. The space of persistent refusal loop conflicts (contradictory cascades where local agent refutations cannot be resolved to a global boundary) is isomorphic to the first cohomology group $H^1(\mathcal{R}^*)$.*

Proof. Let $\omega \in \mathcal{R}^1$ represent local refusal differences. (1) Cocycle condition: ω is a cocycle ($d_1\omega = 0$) if and only if for every 2-simplex (triangle of agents) (v_0, v_1, v_2) , we have $\omega(v_1, v_2) - \omega(v_0, v_2) + \omega(v_0, v_1) = 0$. This requires local consistency around triangles. (2) Coboundary condition: ω is a coboundary ($\omega = d_0\eta$) if and only if the local disagreements can be resolved by assigning a unique refusal state $\eta(v)$ to each agent, such that $\omega(v_0, v_1) = \eta(v_1) - \eta(v_0)$. (3) Obstruction: $H^1(\mathcal{R}^*) = \ker d_1 / \text{im } d_0$ represents the locally consistent but globally irresolvable conflicts. Thus, any non-zero cohomology class $[\omega] \in H^1(\mathcal{R}^*)$ corresponds to a persistent, irresolvable refusal loop conflict. $\square$

3.7 Algebraic Consensus and Relativistic Horizons

In a trillion-agent swarm, physical constraints (relativity and Shannon limits) prohibit global coordination. Consensus must be achieved locally and glued together. Let X be the communication metric space of the agents, and $\mathcal{C}$ be the consensus sheaf on X .

Definition 3.8 (Čech Consensus Complex). Let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X by local communication neighborhoods. The Čech cochain complex $\check{C}^*(\mathcal{U}, \mathcal{C})$ is defined by:

$$\check{C}^p(\mathcal{U}, \mathcal{C}) = \prod_{(i_0, \dots, i_p)} \mathcal{C}(U_{i_0} \cap \dots \cap U_{i_p})$$

with the Čech differential $\delta_p : \check{C}^p(\mathcal{U}, \mathcal{C}) \rightarrow \check{C}^{p+1}(\mathcal{U}, \mathcal{C})$.

Theorem 3.10 (Consensus Sheaf Obstruction). *A family of local consensus states $\{s_i \in \mathcal{C}(U_i)\}_{i \in I}$ can be glued to a unique global consensus state $s \in \mathcal{C}(X)$ if and only if its Čech boundary $\delta s \in \check{C}^1(\mathcal{U}, \mathcal{C})$ is a coboundary, meaning the obstruction class $[\delta s]$ in the first Čech cohomology group $\check{H}^1(\mathcal{U}, \mathcal{C})$ vanishes.*

Proof. Let $s = (s_i)_{i \in I} \in \check{C}^0(\mathcal{U}, \mathcal{C})$ represent the local consensus states. The Čech differential $\delta_0 : \check{C}^0 \rightarrow \check{C}^1$ maps local sections to their pairwise differences: $(\delta_0 s)_{ij} = s_j|_{U_i \cap U_j} - s_i|_{U_i \cap U_j}$. By the sheaf gluing axiom, local sections glue if and only if they agree on all overlaps, which is $(\delta_0 s)_{ij} = 0$. If they disagree, $\omega = \delta_0 s$ is a non-zero 1-cocycle. The class $[\omega]$ vanishes in $\check{H}^1$ if and only if there exists a correction ϵ such that $\omega = \delta_0 \epsilon$. If so, modified local sections $s'_i = s_i - \epsilon_i$ satisfy $\delta_0 s'_i = 0$ and glue uniquely. $\square$

Let $d(x, y)$ be the spatial metric on X , and $v \leq c$ be the signal velocity. A consensus epoch has a bounded duration T .

Theorem 3.11 (Relativistic Consensus Horizon). *If the spatial diameter of the agent network $L = \sup_{x, y \in X} d(x, y)$ satisfies $L > vT$, then the Čech cohomology group $\check{H}^1(\mathcal{U}, \mathcal{C})$ has rank $k \geq \lfloor L/vT \rfloor - 1$. This implies that global consensus is algebraically obstructed, partitioning the swarm into at least $\lfloor L/vT \rfloor$ causally disconnected consensus zones.*

Proof. Let $\mathcal{U} = \{U_i\}_{i \in I}$ be the cover of X where each U_i is a causal cone of radius $r = vT/2$. Since $L > vT$, the network diameter is larger than the diameter of any individual cover element $2r = vT$. Choose points $x_0, \dots, x_N$ in X with $N = \lfloor L/vT \rfloor$ such that $d(x_a, x_{a+1}) > vT$ for all a . Causal cones U_{x_a} and $U_{x_{a+1}}$ do not overlap since $r + r = vT < d(x_a, x_{a+1})$. Thus, the overlap graph of the cover $\mathcal{U}$ has at least N disconnected components. The Čech cochain complex

decomposes as a direct sum, and $\check{H}^1(\mathcal{U}, \mathcal{C})$ contains independent classes for each boundary mismatch. The rank of the cohomology group is at least $N - 1 = \lfloor L/vT \rfloor - 1$. Since the rank is non-zero, global consensus cannot be established within the epoch T . $\square$

Chapter 4

Continuous Limits and Planetary Dynamics

At trillion-agent density, tracking individual agent state transitions and executing discrete retractive gates becomes computationally intractable. By shifting from discrete agents to continuous density fields on a Riemannian manifold, we reformulate admission, manufacture, and receipting as partial differential equations and integral operators.

4.1 The Continuous Limit of the Chatman Equation

Let $N \sim 10^{12}$, and let $\mathcal{M}$ be a compact d -dimensional Riemannian manifold with metric tensor g . The empirical distribution of agents on $\mathcal{M}$ is a probability measure $\rho_N(\mathbf{x}, t) = \frac{1}{N} \sum_{i=1}^N \delta_{\mathbf{x}_i(t)}(\mathbf{x})$. As $N \rightarrow \infty$, ρ_N converges weakly to a continuous, differentiable density field $\rho(\mathbf{x}, t)$ satisfying $\int_{\mathcal{M}} \rho(\mathbf{x}, t) dV_g = 1$, where dV_g is the Riemannian volume form. We define the observation field $\mathcal{O}(\mathbf{x}, t) \in C^0(\mathcal{M})$ as a continuous density of information gathered by the swarm.

We define the continuous admission operator α as a pointwise functional mapping the raw observation field $\mathcal{O}(\mathbf{x}, t)$ to the admitted observation field $\mathcal{O}^*(\mathbf{x}, t)$:

$$\mathcal{O}^*(\mathbf{x}, t) = \alpha[\mathcal{O}](\mathbf{x}, t) = \theta(\mathbf{g}(\mathcal{O}(\mathbf{x}, t))) \cdot \mathcal{O}(\mathbf{x}, t)$$

where $\mathbf{g}$ is the obligation battery, and $\theta(\mathbf{u}) = \prod_{j=1}^m \Theta(u_j)$ represents the joint Heaviside function of the denial monoid boundary. We model the manufacturing process as a non-local integral operator:

$$\mathcal{A}(\mathbf{x}, t) = \mu[\mathcal{O}^*](\mathbf{x}, t) = \int_{\mathcal{M}} \mathcal{K}(\mathbf{x}, \mathbf{y}) \mathcal{O}^*(\mathbf{y}, t) dV_g(\mathbf{y})$$

where $\mathcal{K}(\mathbf{x}, \mathbf{y})$ is a symmetric, compact manufacturing kernel.

Theorem 4.1 (Chatman Mean-Field Convergence). *Let $O_N(d\mathbf{y}, t) = \frac{1}{N} \sum_{i=1}^N o_i(t) \delta_{\mathbf{y}_i(t)}(d\mathbf{y})$ converge weakly to a continuous density $\mathcal{O}(\mathbf{y}, t) dV_g(\mathbf{y})$ as $N \rightarrow \infty$. Under a Lipschitz-continuous manufacturing kernel $\mathcal{K}(\mathbf{x}, \mathbf{y})$ and bounded, continuous obligations $\mathbf{g}$, the sequence of discrete actuation fields $\mathcal{A}_N(\mathbf{x}, t)$ converges strongly in $L^2(\mathcal{M})$ to the continuous actuation field:*

$$\lim_{N \rightarrow \infty} \|\mathcal{A}_N(\mathbf{x}, t) - \mathcal{A}(\mathbf{x}, t)\|_{L^2(\mathcal{M})} = 0.$$

Proof. The difference is: $\mathcal{A}_N(\mathbf{x}, t) - \mathcal{A}(\mathbf{x}, t) = \int_{\mathcal{M}} \mathcal{K}(\mathbf{x}, \mathbf{y}) \alpha[O_N](d\mathbf{y}, t) - \int_{\mathcal{M}} \mathcal{K}(\mathbf{x}, \mathbf{y}) \alpha[\mathcal{O}](\mathbf{y}, t) dV_g(\mathbf{y})$. Because $\mathcal{K}$ is Lipschitz-continuous on the compact manifold $\mathcal{M}$, the integral operator is compact. The weak convergence of $O_N \rightarrow \mathcal{O}$ combined with the continuity of the local admission operator α guarantees that the image under the compact integration operator converges strongly in $L^2(\mathcal{M})$. Thus, the norm of the difference vanishes. $\square$

4.2 Fluid Dynamics of Consequence Conservation

Let $\mathcal{C}(\mathbf{x}, t)$ represent the spatial density of uncommitted actuation consequences in transit. The transport of consequence is governed by advection (velocity field $\mathbf{v}(\mathbf{x}, t)$) and diffusion (tensor $\mathbf{D}(\mathbf{x}, t)$), defining the flux $\mathbf{J}_C = \mathcal{C}\mathbf{v} - \mathbf{D}\nabla\mathcal{C}$. The consequence density satisfies the advection-diffusion-reaction equation:

$$\frac{\partial \mathcal{C}}{\partial t} + \nabla \cdot (\mathcal{C}\mathbf{v}) = \nabla \cdot (\mathbf{D}\nabla\mathcal{C}) + \mathcal{S}_C - \mathcal{R}_C$$

where the source is $\mathcal{S}_C = \sigma(\mathbf{x}, t)\mathcal{O}^*(\mathbf{x}, t)$ and the sink is $\mathcal{R}_C = \gamma(\mathbf{x}, t)\mathcal{C}(\mathbf{x}, t)$.

Theorem 4.2 (Global Consequence Conservation). *On a compact Riemannian manifold $\mathcal{M}$ without boundary ($\partial\mathcal{M} = \emptyset$), the total uncommitted consequence $C_{\text{total}}(t) = \int_{\mathcal{M}} \mathcal{C}(\mathbf{x}, t) dV_g$ satisfies:*

$$\frac{d}{dt} C_{\text{total}}(t) = \int_{\mathcal{M}} \sigma(\mathbf{x}, t)\mathcal{O}^*(\mathbf{x}, t) dV_g - \int_{\mathcal{M}} \gamma(\mathbf{x}, t)\mathcal{C}(\mathbf{x}, t) dV_g.$$

Furthermore, the total consequence committed to the ledger $C_{\text{ledger}}(t)$ satisfies:

$$C_{\text{ledger}}(t) = C_{\text{ledger}}(0) + \int_0^t \int_{\mathcal{M}} \gamma(\mathbf{x}, s)\mathcal{C}(\mathbf{x}, s) dV_g ds$$

ensuring the total system consequence $C_{\text{total}}(t) + C_{\text{ledger}}(t)$ is conserved.

Proof. Integrating the transport equation over $\mathcal{M}$ yields $\int_{\mathcal{M}} \frac{\partial \mathcal{C}}{\partial t} dV_g + \int_{\mathcal{M}} \nabla \cdot \mathbf{J}_C dV_g = \int_{\mathcal{M}} (\mathcal{S}_C - \mathcal{R}_C) dV_g$. By the Divergence Theorem, since $\partial\mathcal{M} = \emptyset$, the integral of the divergence of the flux vanishes. Leibniz's rule allows pulling the time derivative out of the spatial integral. Integrating this relation from 0 to t simplifies to the global conservation law. $\square$

4.3 Dynamic System Limits and Feedback Loop Stability

At planetary scale, the propagation speed of information is bounded by the speed of light in the communication medium: $v_{\text{prop}} = \beta c$ (typically $\beta \approx 0.67$ for fiber optics). For a planet of diameter D , the maximum propagation delay is $\tau_{\text{max}} = D/v_{\text{prop}}$. Let local state deviations $s(\mathbf{x}, t)$ satisfy:

$$\frac{\partial s(\mathbf{x}, t)}{\partial t} = -\omega_0 s(\mathbf{x}, t) - K_f \int_{\mathcal{M}} w(\mathbf{x}, \mathbf{y}) s(\mathbf{y}, t - \tau(\mathbf{x}, \mathbf{y})) dV_g(\mathbf{y}) + u(\mathbf{x}, t)$$

where $\omega_0 > 0$ is the relaxation rate, K_f is feedback gain, and $\tau(\mathbf{x}, \mathbf{y})$ geodesic delay.

Theorem 4.3 (Planetary Delay-Gain Stability Bound). *For a planetary-scale swarm feedback loop with maximum propagation delay $\tau_{\max}$, the system is asymptotically stable if and only if the global feedback gain K_f satisfies:*

$$K_f < \sqrt{\omega_0^2 + \omega_c^2}$$

where the critical crossover frequency ω_c is the unique solution to the transcendental equation $\omega_c = \omega_0 \tan(\omega_c \tau_{\max})$. In the limit of negligible local damping ($\omega_0 \rightarrow 0$), the stability boundary simplifies to:

$$K_f < \frac{\pi}{2\tau_{\max}}.$$

Proof. Taking the Fourier transform, the characteristic equation of the system is $\lambda + \omega_0 + K_f e^{-\lambda \tau_{\max}} = 0$. Setting $\lambda = i\omega_c$ for marginal stability yields $i\omega_c + \omega_0 + K_f (\cos(\omega_c \tau_{\max}) - i \sin(\omega_c \tau_{\max})) = 0$. Separating real and imaginary parts: $\cos(\omega_c \tau_{\max}) = -\frac{\omega_0}{K_f}$ and $\sin(\omega_c \tau_{\max}) = \frac{\omega_c}{K_f}$. Squaring and adding both equations yields $K_f = \sqrt{\omega_0^2 + \omega_c^2}$. Dividing the sine equation by the cosine equation gives the relation for the critical frequency. When $\omega_0 \rightarrow 0$, the cosine equation requires $\cos(\omega_c \tau_{\max}) = 0$, giving the lowest positive frequency as $\omega_c = \frac{\pi}{2\tau_{\max}}$ and critical gain $K_f = \omega_c = \frac{\pi}{2\tau_{\max}}$. $\square$

Theorem 4.4 (Consensus Latency-Stability Inequality). *To prevent consensus forks and maintain the integrity of the B_4 conformance invariant in a planetary-scale swarm, the loop control frequency f_{loop} must satisfy:*

$$f_{\text{loop}} \leq \frac{1}{2T_{\text{consensus}}} \leq \frac{v_{\text{prop}}}{2D}.$$

Proof. The consensus delay introduces a phase lag $\theta_{\text{consensus}} = 2\pi f_{\text{loop}} T_{\text{consensus}}$ in the verification feedback loop. If $\theta_{\text{consensus}} > \pi$, the feedback shifts from negative to positive, triggering a spatial bifurcation where agents act on uncommitted, out-of-date states, violating conformance ($x \notin \mathcal{P}$). Thus, stability requires $\theta_{\text{consensus}} \leq \pi$, which yields $f_{\text{loop}} \leq \frac{1}{2T_{\text{consensus}}}$. $\square$

For a terrestrial swarm ($D \approx 12,742$ km) communicating via standard fiber optic networks ($\beta \approx 0.67$), we have $v_{\text{prop}} \approx 2 \times 10^8$ m/s, yielding $\tau_{\max} \approx 63.7$ ms. The maximum stable control frequency is $f_{\text{control}} \leq 7.85$ Hz. This limits the actuation feedback loops of trillion-agent planetary swarms to sub-10 Hz regimes, proving that local sub-swarms must operate with decoupled, localized sub-retractions to avoid planetary-scale instability.

Chapter 5

The Agent Byte and Branchless Fleet Admission

We now scale from one manufacture to a population. The keystone’s Construction projected an agent onto eight bits; here we build that byte from the running **DenialPolarity** lanes, prove that fleet admission is branchless mask algebra gating 64 agents per word per lane, and compute the footprint and sweep time of a 10^{10} -agent population exactly (footprint) and by estimate (time).

5.1 The agent8 byte from the denial lanes

The **praxis** refusal machinery exposes a denial word with eight lanes: the **ADMITTED** lane and seven denial lanes (**WATCHDOG_DRAINED**, **PRECONDITION_FAILED**, **SLA_BREACH**, **AUTHORIZATION_DENIED**, **RESOURCE_EXHAUSTED**, **OBJECT_LIFECYCLE_VIOLATION**, **CONFORMANCE_GATE_FAILED**), each mapping to a bucket of the eight-category taxonomy of **refusal.rs**. The agent byte is the per-agent projection of these lanes into a positive (“OK”) polarity.

Construction 5.1 (The agent8 byte). Project each agent a to a status byte $b_a \in \{0, 1\}^8$ whose lanes are the OK-polarity complements of the denial lanes:

$$b_a = (\text{admitted}, \text{evidence}, \text{budget}, \text{authority}, \text{healthy}, \text{conformant}, \text{receipted}, \text{replayable}),$$

with lane value 1 meaning the corresponding obligation holds and 0 meaning its denial lane is set. Write $\mathbf{1} = 0xFF$ for the all-OK byte. An agent is *admissible* iff $b_a = \mathbf{1}$, equivalently iff its 8-lane denial restriction is $\mathbf{0}$.

Remark. As in the keystone, the **receipted** and **replayable** lanes are set only by chained receipts whose frames commit the law-bearing fields; the byte is therefore a faithful projection of the agent’s receipted lifecycle, not a self-report. A peer with context κ reads a counterpart’s entire lawful status in one byte-load.

5.2 Two layouts and the branchless admission sweep

A fleet of N agents is a matrix of bits: N agents $\times$ 8 lanes. Two storage layouts matter.

Definition 5.1 (AoS and SoA layouts). The *array-of-structures* (AoS) layout stores the 8-bit byte of each agent contiguously; a 64-bit machine word holds 8 agents' bytes. The *structure-of-arrays* (SoA), or *bit-plane*, layout stores each lane as its own bitset: lane ℓ is a bit-array $P_\ell \in \{0, 1\}^N$, and a 64-bit word of P_ℓ holds the lane- ℓ bit of 64 agents.

Theorem 5.1 (Branchless admission gates 64 agents per word). *Under the SoA layout of Definition 5.1, the admissibility of 64 agents is computed by the single branchless word expression:*

$$\text{adm} = P_1 \& P_2 \& P_3 \& P_4 \& P_5 \& P_6 \& P_7 \& P_8,$$

where each P_ℓ here denotes the 64-agent word of plane ℓ and $\&$ is bitwise AND. Bit j of adm is 1 iff agent j (of the 64) has all eight lanes OK, i.e. iff $b_{a_j} = \mathbf{1}$. The computation uses exactly 7 AND instructions, contains no data-dependent branch, and processes 64 agents; its cost is $7/64 \approx 0.11$ instructions per agent, independent of the lane values.

Proof. Agent a_j is admissible iff $\bigwedge_{\ell=1}^8 b_{a_j \ell} = 1$ (Construction 5.1). Bit j of P_ℓ equals $b_{a_j \ell}$ by the SoA definition. Bitwise AND acts coordinatewise, so bit j of $P_1 \& \dots \& P_8$ equals $\bigwedge_{\ell} b_{a_j \ell}$, which is 1 iff a_j is admissible. A left-fold of AND over 8 operands is 7 binary ANDs. AND is a straight-line instruction with no branch, so the sequence is branchless; it reads 8 words (64 agents' worth of each lane) and writes one, regardless of the bits' values. The per-agent instruction count is $7/64$. $\square$

Corollary 5.2 (Reduction to a single denial word). *Equivalently, in denial polarity, let D_ℓ be the 64-agent word of denial lane ℓ ($D_\ell = \neg P_\ell$). Then the denied set is $\bigvee_{\ell} D_\ell$ and the admitted set its complement, matching the join-semilattice of the denial monoid: fleet admission is the word-parallel join of denial lanes followed by one complement. Adding a ninth lane costs one more AND (resp. OR) per 64 agents.*

Proof. By Theorem 5.1, the admissibility of 64 agents is given by the word expression $\text{adm} = \bigwedge_{\ell=1}^8 P_\ell$. In denial polarity, each OK-polarity bit-plane P_ℓ is related to its denial-polarity counterpart D_ℓ by $P_\ell = \neg D_\ell$. Applying De Morgan's laws coordinatewise over the bitwise operators yields:

$$\text{adm} = \bigwedge_{\ell=1}^8 (\neg D_\ell) = \neg \left(\bigvee_{\ell=1}^8 D_\ell \right).$$

Let $\text{den} = \bigvee_{\ell=1}^8 D_\ell$ be the denied set. The fleet admission condition is therefore $\text{adm} = \neg \text{den}$, which corresponds to the join-semilattice composition of the denial monoid $(D, \vee, \mathbf{0})$ under the isomorphism $D_n \cong 2^{[n]}$.

Computing den requires 7 bitwise OR instructions to left-fold the 8 lanes, and 1 bitwise NOT to compute the complement, totaling 8 instructions. Adding a ninth lane expands the join to $\bigvee_{\ell=1}^9 D_\ell$, requiring 8 OR instructions and 1 NOT, which increases the cost by exactly one instruction per 64 agents. $\square$

Remark. Theorem 5.1 is stated for a 64-bit scalar word. A 512-bit SIMD register (AVX-512) executes the same 7 ANDs over 512 agents per lane, dropping the per-agent cost to $7/512 \approx 0.014$ instructions/agent. The algebra is unchanged; only the word widens. This is the same mechanism by which the vector JSON parser classifies 64 bytes per instruction — admission is byte-classification lifted to agents.

Construction 5.2 (Borrow-Safe Zero-Lane Detector). Let $w \in \{0,1\}^{64}$ be a denial word comprising eight 8-bit lanes. Define the lane masks $L_{\text{high}} = 0x8080_8080_8080$ and $L_{\text{low7}} = 0x7f7f_7f7f_7f7f_7f7f$. The zero-lane bitmask $z(w) \in \{0,1\}^{64}$ is computed branchlessly by:

$$z(w) = \neg \left(((w \& L_{\text{low7}}) + L_{\text{low7}}) \vee w \right) \& L_{\text{high}}, \quad (5.1)$$

where $+$ is wrapping addition and $\vee, \&, \neg$ are bitwise operations. Bit 7 of lane j in $z(w)$ is 1 iff lane j in w is exactly $0x00$, and 0 otherwise.

Theorem 5.3 (SWAR Fleet Sweep Verification). *Let w be a 64-bit word packing 8 agents. If w is gated against the broadcast required mask, the number of admitted agents is obtained by the popcount:*

$$\text{admitted}(w) = \text{popcount}(z(w)).$$

This sweeps 8 agents in one CPU instruction with zero branches and no lane borrow leakage.

Proof. We analyze the branchless zero-lane detector expression on a single 8-bit lane (byte) $b \in \{0,1\}^8$. Let $m = 0x7f$ and $h = 0x80$. The byte-level restriction of equation (5.1) is: $z(b) = \neg \left(((b \& m) + m) \vee b \right) \& h$. We evaluate this for two cases:

Case 1: $b = 0x00$. We have $b \& m = 0x00$, so the addition gives $0x00 + 0x7f = 0x7f$. Then $0x7f \vee 0x00 = 0x7f$. Taking the bitwise NOT yields $\neg(0x7f) = 0x80$. Masking with h gives $0x80 \& 0x80 = 0x80$. Thus, the high bit of $z(b)$ is 1.

Case 2: $b \neq 0x00$. Subcase 2a: The low 7 bits of b are not all 0. Then $b \& m \geq 0x01$, so the addition is $\geq 0x01 + 0x7f = 0x80$. Thus, the high bit of $((b \& m) + m)$ is 1, which forces the high bit of $((b \& m) + m) \vee b$ to be 1. Taking the bitwise NOT forces the high bit of $z(b)$ to be 0. Subcase 2b: The low 7 bits of b are all 0, but the high bit is 1 (i.e., $b = 0x80$). Then $b \& m = 0x00$, so the addition is $0x7f$. The join is $0x7f \vee 0x80 = 0xff$. The bitwise NOT is $\neg(0xff) = 0x00$, so the high bit of $z(b)$ is 0.

Thus, the high bit of lane j in $z(w)$ is 1 if and only if lane j of w is exactly $0x00$. Since the maximum value of $w \& L_{\text{low7}}$ in any byte is $0x7f$ and we add $L_{\text{low7}} = 0x7f7f\dots$, the addition within each byte is at most $0x7f + 0x7f = 0xfe$, which never produces a carry into the next byte (no borrow/carry leakage). The number of admitted agents (fully zero lanes) is the number of set high bits, which is counted by $\text{popcount}(z(w))$ in a single instruction on scalar processors. $\square$

5.3 Footprint and sweep time of a population

Proposition 5.4 (Population footprint is exact). *A population of $N = 10^{10}$ agents, each an 8-bit byte (Construction 5.1), occupies:*

$$N \times 8 \text{ bits} = N \text{ bytes} = 10^{10} \text{ bytes} = 10 \text{ GB},$$

in either layout. This is an exact arithmetic identity, not an estimate.

Proof. Eight bits is one byte; N agents is N bytes; 10^{10} bytes = 10 GB under the decimal-GB convention. Both layouts store the same $8N$ bits, hence the same N bytes. No estimated quantity appears. $\square$

Estimate 5.1 (One-pass sweep time is memory-bandwidth-bound). A full admission sweep reads all 8 planes once (10 GB total) and, by Theorem 5.1, performs $7N/64 \approx 1.1 \times 10^9$ AND instructions for $N = 10^{10}$. At scalar issue rates ($\sim 10^{10}$ simple ops/s/core) the compute is ~ 0.1 s single-core and far less across cores and SIMD; at streaming memory bandwidth $B \sim 10^2\text{--}4 \times 10^2$ GB/s the read of 10 GB takes:

$$t_{\text{sweep}} \approx \frac{10 \text{ GB}}{B} \approx 25\text{--}100 \text{ ms}.$$

Compute is dominated by memory traffic: the sweep is *bandwidth-bound*, not compute-bound. The instruction count is exact (Theorem 5.1); the wall-time is an order-of-magnitude estimate from commodity bandwidth, labelled as such.

Corollary 5.5 (Admission throughput). *From Estimate 5.1, a single commodity server admits $N = 10^{10}$ agents in $\sim 25\text{--}100$ ms, i.e. at $\sim 10^{11}\text{--}4 \times 10^{11}$ agent-admission decisions per second per server, bandwidth-bound. This is the supply figure the planetary economics of Chapter 6 weigh against the comprehension-based alternative.*

Proof. By Estimate 5.1, the memory swept for $N = 10^{10}$ agents is 10 GB. Given a commodity memory bus bandwidth $B \in [100, 400]$ GB/s, the time required to stream 10 GB of data into CPU cache is:

$$t_{\text{sweep}} \approx \frac{10 \text{ GB}}{B} \in \left[\frac{10}{400}, \frac{10}{100} \right] \text{ s} = [25, 100] \text{ ms}.$$

The admission throughput T is the number of agent decisions per unit time:

$$T = \frac{N}{t_{\text{sweep}}} \in \left[\frac{10^{10}}{0.1}, \frac{10^{10}}{0.025} \right] = [10^{11}, 4 \times 10^{11}] \text{ decisions/s}.$$

Since compute requires $7N/64 \approx 1.1 \times 10^9$ AND instructions, which executes in ≈ 100 ms on a single 10 GHz core and scales down linearly with multiple cores and vector instructions, the execution time is strictly dominated by the memory read transit time, making the throughput bandwidth-bound. $\square$

Chapter 6

Planetary Capacity and the Economics of Admission

We now weigh demand against supply for the two admission regimes — *bit-parallel* (the sweep) and *comprehension-based* (an LLM or human reading each observation) — and show the gap is not a matter of optimization but of feasibility. All figures in this chapter are Fermi estimates and are labelled; the *ordering* they establish is robust to an order of magnitude in any input.

6.1 Demand

Estimate 6.1 (Planetary admission demand). Take a population of $N \in [10^9, 10^{12}]$ agents (the keystone’s fleet), each re-admitted — on receiving a peer message, on a lifecycle transition, on a watchdog tick — at a rate $\rho \in [1, 10^2] \text{ s}^{-1}$. The aggregate admission demand is:

$$\Lambda_{\text{demand}} = N\rho \in [10^9, 10^{14}] \text{ decisions/s},$$

with a central estimate of $\sim 10^{11}$ – 10^{12} decisions/s for $N \sim 10^{10}$, $\rho \sim 10$.

6.2 Supply, two regimes

Estimate 6.2 (Comprehension-based supply ceiling). A comprehension-based admission decision — reading the observation with an LLM and judging its meaning — costs on the order of 10^{-2} – 10^0 s of accelerator time per decision (the keystone’s $\sim 10^{-2} \text{ s}$ figure), i.e. a per-accelerator rate of ~ 1 – 10^2 decisions/s. Even a planetary datacenter fleet of $\sim 10^6$ – 10^7 accelerators yields a comprehension-supply ceiling:

$$\Lambda_{\text{comp}} \lesssim 10^7\text{--}10^9 \text{ decisions/s},$$

with a conservative central estimate of $\sim 10^8$ decisions/s.

Estimate 6.3 (Bit-parallel supply). By Corollary 5.5, one commodity server supplies $\sim 10^{11}$ decisions/s bandwidth-bound. A modest rack of $\sim 10^2$ servers supplies $\sim 10^{13}$ decisions/s; the primitive is $\sim 10^7 \times$ cheaper per decision than an LLM judgment, so bit-parallel supply scales into the demand band of Estimate 6.1 on hardware of a single facility.

Theorem 6.1 (Affordability separation). *Let Λ_{demand} , Λ_{comp} , Λ_{bit} be the demand and the two supplies. Under the estimates above,*

$$\Lambda_{\text{comp}} \sim 10^8 \ll \Lambda_{\text{demand}} \sim 10^{11}\text{--}10^{12} \lesssim \Lambda_{\text{bit}} \sim 10^{11}\text{--}10^{13}.$$

Hence comprehension-based admission is under-provisioned by three to four orders of magnitude for a 10^{10} -agent fleet, while bit-parallel admission meets demand on a single facility. The separation persists under an order-of-magnitude perturbation of any single input.

Estimate. Substitute the central estimates of Estimates 6.1–6.3. The comprehension ceiling $\Lambda_{\text{comp}} \sim 10^8$ falls short of demand $\sim 10^{11}\text{--}10^{12}$ by $10^3\text{--}10^4$; the bit-parallel supply meets or exceeds it. The gap between the two supplies is the keystone’s per-decision cost ratio $\sim 10^7$ scaled by population. A tenfold error in any one input moves the inequality by at most one decade and does not reverse the three-to-four decade comprehension shortfall. $\square$

Corollary 6.2 (Only bit-parallel admission is affordable). *A planetary control plane that re-admits its population continuously cannot be built on comprehension: no attainable accelerator fleet closes the demand gap. It can be built on the sweep of Chapter 5, whose cost is dominated by a single pass over 10 GB of memory. Comprehension is reserved, as in the keystone, for sampled audit at honest $O(n)$ replay cost — never for the per-message path.*

Proof. By Theorem 6.1, the maximum planetary supply of comprehension-based admission decisions is bounded by $\Lambda_{\text{comp}} \leq 10^9$ decisions/s. For a planetary population $N \geq 10^{10}$ agents re-admitted at rate $\rho \geq 10 \text{ s}^{-1}$, the demand is $\Lambda_{\text{demand}} \geq 10^{11}$ decisions/s. The deficit is: $\Lambda_{\text{demand}} - \Lambda_{\text{comp}} \geq 9.9 \times 10^{10}$ decisions/s, which is a shortfall of 3 to 4 orders of magnitude. In contrast, a single commodity server supplies $\sim 10^{11}$ decisions/s (Corollary 5.5), and a rack of 100 servers supplies $\Lambda_{\text{bit}} \sim 10^{13}$ decisions/s, fully covering the demand. Because the cost of the sweep scales only with the memory streaming of 10 GB, it is within planetary physical and thermodynamic energy budgets. Thus, continuous admission can only be built on the bit-parallel sweep, restricting comprehension to out-of-band sampled audits. $\square$

6.3 The true floor is bandwidth, and it is generous

Remark. Because the sweep is bandwidth-bound (Estimate 5.1), the limiting limit on planetary admission is memory-system throughput, not logic. Bit-plane admission sits comfortably above the Landauer thermodynamic floor and orders of magnitude below the energy of a single comprehension decision; the physical budget of planetary admission is the cost of streaming a few tens of gigabytes, which any facility affords many times per second. Comprehension-based admission has no comparable floor because its cost scales with interior dimension per decision, exactly the quantity the projection principle exists to avoid paying.

Chapter 7

The Geometry of Swarm Conformance

At 10^{12} agents, discrete tracking of individual agent Parikh vectors $\mathbf{x} \in \mathbb{Z}_{\geq 0}^p$ becomes computationally intractable. Conformance checking must transition from discrete combinatorial search to continuous and topological field theories.

7.1 Infinite-Dimensional Trace Geometry and Farkas' Lemma

Let $\mathcal{X}$ be a Banach space representing the space of continuous execution traces (e.g., the space of signed Radon measures $\mathcal{M}(\Omega)$ over the action domain Ω), and let $K \subset \mathcal{X}$ be a closed convex cone representing lawful execution densities. Let $\mathcal{Y}$ be a Banach space representing the continuous marking configuration (state) space. We generalize the discrete net incidence matrix $\mathbf{N}$ to a bounded, continuous linear operator $\mathcal{N} : \mathcal{X} \rightarrow \mathcal{Y}$. The continuous state equation governing the transition is $m^\dagger = m_0 + \mathcal{N}\mu$ for some $\mu \in K$, where $\Delta m = m^\dagger - m_0$ is the marking density transition. The continuous lawful trace region is $\mathcal{P}_{\text{lawful}} = m_0 + \mathcal{N}(K) \subset \mathcal{Y}$. Conformance corresponds to the membership query $\Delta m \in \overline{\mathcal{N}(K)}$.

Theorem 7.1 (Infinite-Dimensional Farkas Separation). *Let $\mathcal{X}$ and $\mathcal{Y}$ be locally convex topological vector spaces, $K \subset \mathcal{X}$ a closed convex cone, and $\mathcal{N} : \mathcal{X} \rightarrow \mathcal{Y}$ a continuous linear operator. Fix $\Delta m \in \mathcal{Y}$. Exactly one of the following holds:*

1. $\Delta m \in \overline{\mathcal{N}(K)}$ (the continuous execution relaxation is feasible).
2. There exists a continuous linear functional $y \in \mathcal{Y}^*$ (the dual potential field) such that $\mathcal{N}^*y \leq 0$ (i.e., $\langle \mathcal{N}^*y, \mu \rangle \leq 0$ for all $\mu \in K$) and $\langle y, \Delta m \rangle > 0$, where $\mathcal{N}^* : \mathcal{Y}^* \rightarrow \mathcal{X}^*$ is the adjoint operator.

Proof. Since $\overline{\mathcal{N}(K)}$ is a closed convex cone in the locally convex space $\mathcal{Y}$, and the point $\Delta m \notin \overline{\mathcal{N}(K)}$, the Hahn-Banach Separation Theorem [9] guarantees the existence of a continuous linear functional $y \in \mathcal{Y}^*$ and a constant $\alpha \in \mathbb{R}$ such that $\langle y, z \rangle \leq \alpha < \langle y, \Delta m \rangle$ for all $z \in \overline{\mathcal{N}(K)}$. Since $0 \in \overline{\mathcal{N}(K)}$, we have $0 \leq \alpha$, which implies $\langle y, \Delta m \rangle > 0$. Suppose there exists some $z_0 \in \overline{\mathcal{N}(K)}$ such that $\langle y, z_0 \rangle > 0$. Because $\overline{\mathcal{N}(K)}$ is a cone, $\lambda z_0 \in \overline{\mathcal{N}(K)}$ for all $\lambda > 0$. Taking the limit as $\lambda \rightarrow \infty$ yields $\langle y, \lambda z_0 \rangle \rightarrow \infty$, which violates the upper bound α . Thus, we must have $\langle y, z \rangle \leq 0$ for all $z \in \overline{\mathcal{N}(K)}$, and setting $\alpha = 0$ preserves the inequality. For any $\mu \in K$,

we have $\mathcal{N}\mu \in \mathcal{N}(K) \subset \overline{\mathcal{N}(K)}$. Therefore: $\langle y, \mathcal{N}\mu \rangle = \langle \mathcal{N}^*y, \mu \rangle \leq 0$. The continuous functional $y \in \mathcal{Y}^*$ serves as the separating hyperplane, certifying nonconformance. $\square$

7.2 Differential Geometry of Swarm Action Fields

We treat the collective action space as a smooth Riemannian manifold $\mathcal{M}$ with metric tensor g . The spatial distribution of the swarm is represented as a probability density $\rho(t, x) \in \mathcal{P}(\mathcal{M})$ satisfying $\int_{\mathcal{M}} \rho(t, x) dV_g = 1$, flowing along action fields $V(t, x) \in T_x\mathcal{M}$ governed by the continuity equation $\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho V) = 0$. The minimum-energy path transitioning the swarm from initial ρ_0 to target ρ_f defines the Wasserstein distance $W_2(\rho_0, \rho_f)$ in $\mathcal{P}_2(\mathcal{M})$:

$$W_2^2(\rho_0, \rho_f) = \inf_{\rho, V} \left\{ \int_0^1 \int_{\mathcal{M}} g(V(t, x), V(t, x)) \rho(t, x) dV_g dt \right\}$$

subject to $\partial_t \rho + \nabla \cdot (\rho V) = 0$ with $\rho(0, \cdot) = \rho_0$ and $\rho(1, \cdot) = \rho_f$. Under this formulation, the swarm's cost-optimal execution corresponds to a geodesic in the Wasserstein space $\mathcal{P}_2(\mathcal{M})$.

To enforce conformance constraints, we introduce a potential barrier $U : \mathcal{M} \rightarrow \mathbb{R} \cup \{+\infty\}$, where unauthorized regions correspond to $U(x) = +\infty$. Incorporating agent exploration entropy (diffusion coefficient $\sigma > 0$), the swarm's gradient flow matches the Fokker-Planck equation on the manifold: $\frac{\partial \rho}{\partial t} = \nabla \cdot (\rho \nabla U) + \sigma \Delta_g \rho$, where Δ_g is the Laplace-Beltrami operator. The stability of the swarm's coordination is governed by the Riemann curvature tensor R_{ijk}^l of the action manifold. Negative sectional curvature drives exponential divergence of trajectories, while positive sectional curvature focalizes agent flows along coordinate geodesics.

7.3 Topological Conformance Bounds in Exponential Token Games

Concurrent state spaces suffer from exponential state-space explosion. We bypass this limit by bounding conformance verification using topological invariants of the execution space. We model the concurrent state space as a cubical complex $\mathbb{K}$, where vertices represent local state configurations and directed edges represent transitions. Let $\mathbb{K}_{lawful} \subset \mathbb{K}$ represent the sub-complex of lawful markings.

Theorem 7.2 (Homological Conformance Bound). *Let $\mathbb{K}_{lawful}$ be the lawful execution sub-complex. If there exists a deformation retraction $r : \mathbb{K}_{lawful} \rightarrow \mathcal{B}$ onto a lower-dimensional boundary skeleton $\mathcal{B}$, then any execution trace $\gamma \subset \mathbb{K}_{lawful}$ can be projected to a path $r \circ \gamma \subset \mathcal{B}$. Furthermore, let $H_n(\mathbb{K}_{lawful})$ be the n -th homology group of the lawful complex. The verification complexity of conformance is bounded by the dimension of the homology groups:*

$$\dim H_1(\mathbb{K}_{lawful}) = b_1(\mathbb{K}_{lawful})$$

where b_1 is the first Betti number, rather than the exponential cardinality of the state space $|\mathbb{K}|$.

Proof. Let $\gamma_1, \gamma_2 : [0, 1] \rightarrow \mathbb{K}_{lawful}$ be two directed execution paths sharing start and end points. If they are homologous, there exists a 2-chain σ in $\mathbb{K}_{lawful}$ such that $\partial \sigma = \gamma_1 - \gamma_2$. Since the boundary operator ∂ preserves the topological boundaries of the lawful space, the region between the paths contains no "holes" corresponding to unauthorized states or deadlocks.

The projection under the retraction r maps the 2-chain σ to a 2-chain $r(\sigma)$ in the boundary skeleton $\mathcal{B}$. The verification of conformance is thus reduced to checking membership of the class $[r(\gamma)] \in H_1(\mathcal{B})$. The size of the certificate required to prove conformance is proportional to the Betti number $b_1(\mathbb{K}_{lawful})$ and the dimension of $\mathcal{B}$, isolating the verification boundary to $O(\kappa)$ coordinates. $\square$

Chapter 8

Scale Verification and Reality Addressing

8.1 Swarm Scaling and Hierarchical Zero-Defect Bounds

We define a **Hierarchical Merkle Rollup** over the swarm.

Definition 8.1 (Member Record). For each agent $a \in [N]$, its local execution is projected into a member record:

$$r_a = \langle a, b_a, h_{a,\text{term}}, \text{refusal} \rangle$$

where $b_a \in \mathbb{B}^8$ is the status byte, and the terminal hash $h_{a,\text{term}} \in \{0, 1\}^{256}$ is defined as:

$$h_{a,\text{term}} = \begin{cases} \Psi(a), & \text{if } b_a \ \& \ \text{A_ADMITTED} \neq 0, \\ \text{H}(\text{refusal}), & \text{if } b_a \ \& \ \text{H_HALTED} \neq 0. \end{cases}$$

Here, $\Psi(a)$ represents the deterministic commitment to the agent's admission receipt.

Definition 8.2 (Group Replay Root). Let the swarm be partitioned into G disjoint groups of size $S = N/G$. For each group $g \in [G]$, its replay root $h_{g,\text{root}}$ is computed as:

$$h_{g,\text{root}} = \text{H}(\text{sort}(\{h_{a,\text{term}} : a \in g\})).$$

Sorting terminal hashes ensures that the replay root is invariant under member execution order.

Definition 8.3 (Cell Hash). A cell receipt composes G group replay roots into a rolling commitment:

$$h_{\text{cell},i} = \text{fold_event}(h_{\text{cell},i-1}, h_{g_i,\text{root}}) \quad \forall i \in [1, G].$$

Definition 8.4 (Supra-Cell Hash). Multiple cells are aggregated into a supra-cell summary:

$$h_{\text{supra},j} = \text{fold_event}(h_{\text{supra},j-1}, \text{summary_line}(s_j)) \quad \forall j \in [1, C]$$

where $\text{summary_line}(s_j) = j \parallel n_j \parallel \text{admitted}_j \parallel \text{refused}_j \parallel h_{\text{cell},j}$.

Theorem 8.1 (Zero-Defect Verification Complexity). *The computational complexity for a verifier to validate the integrity of a supra-cell receipt is $O(C \cdot G)$, independent of the total agent count $N = 10^{12}$.*

Proof. To verify the supra-cell receipt, the verifier reads the C summaries and recomputes the cell hashes $h_{\text{cell},j}$ using the pre-computed group roots $h_{g_i,\text{root}}$. Refolding the supra-cell hash requires C summary line hashes, and recomputing the cell hashes requires $C \cdot G$ event foldings. Thus, the total operations are $O(C \cdot G)$, which is independent of the swarm size N . $\square$

Theorem 8.2 (Anti-Cheat Soundness). *Let k be the number of agent execution states altered by an adversary. The probability $\varepsilon_{\text{cheat}}$ that the adversary can mutate these states without changing the supra-cell hash h_{supra} is bounded by:*

$$\varepsilon_{\text{cheat}} \leq (k + G + C) \cdot \varepsilon(\lambda)$$

where $\varepsilon(\lambda) \approx 2^{-128}$ is the collision probability of BLAKE3.

Proof. Any mutation in an agent's status byte b_a or terminal hash $h_{a,\text{term}}$ alters the set $\{h_{a,\text{term}} : a \in g\}$. Under the collision resistance of H, this mutates the group replay root $h_{g,\text{root}}$ except with probability $\varepsilon(\lambda)$. By induction, this alters the cell hash $h_{\text{cell},j}$ and the summary line, changing the terminal supra-cell hash h_{supra} unless a collision is found. Summing the independent collision bounds at the member, group, and cell layers yields the theorem. $\square$

An individual agent a can be verified on challenge in $O(|S_{\text{trace}}| + S_{\text{group}})$ time by replaying only its group's local members.

8.2 Localized Replay Geometries

Let the swarm execution be represented as a directed acyclic graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ is the set of virtual state transitions (agent lifecycles) and $\mathcal{E}$ represents causal dependencies (token/data flows).

Theorem 8.3 (Localized Replay Decoupling). *Let $S \subset \mathcal{V}$ be a sub-region of interest. If the states of the incoming boundary nodes:*

$$\partial^- S = \{u \in \mathcal{V} \setminus S : \exists v \in S, (u, v) \in \mathcal{E}\}$$

are cryptographically signed and committed, then the validity of all transitions in S can be verified with complexity $O(|S| + |\partial^- S|)$, independent of $|\mathcal{V} \setminus S|$.

Proof. Let f_v be the deterministic transition function of node $v \in S$. The inputs to f_v are the outputs of its parent nodes. By topological sorting of S , every input to a node $v \in S$ is either an output of another node $u \in S$ or an output of a boundary node $w \in \partial^- S$. Since the outputs of $\partial^- S$ are committed and authenticated, they serve as trusted oracle inputs. The verifier recomputes the transitions f_v sequentially for all $v \in S$, requiring exactly $|S|$ evaluations of local transition functions and $|\partial^- S|$ boundary lookups, yielding a complexity of $O(|S| + |\partial^- S|)$. $\square$

8.3 Virtual State Transitions and Physical Reality Addressing

To ground traces, we map high-cardinality virtual state transitions to the physical universe.

Definition 8.5 (Reality Address Record). A virtual transition $v \in \mathcal{V}$ is bound to physical reality by a Reality Address Record:

$$R_v = \langle s_v, t_v, w_v, p_v \rangle$$

where s_v is the IRI of v , and the anchors are extracted from the admitted graph triples:

- t_v : ‘inXSDDateTimeStamp’ (‘OWL-Time’) representing the physical time of execution.
- w_v : ‘asWKT’ (‘GeoSPARQL’) representing the physical Well-Known Text spatial coordinate.
- p_v : ‘wasAttributedTo’ (‘PROV-O’) representing the physical entity or hardware agent.

Theorem 8.4 (Anchor Sufficiency and Refusal). *Let β_{reality} be the reality addressing map. If a subject s contains no public ontology coordinates, the mapping retracts to the refusal constant:*

$$t_v = \emptyset \wedge w_v = \emptyset \wedge p_v = \emptyset \implies \beta_{\text{reality}}(v) = \perp_{\text{reality}}$$

which propagates as a ‘RealityAddressIllFormed’ refusal.

Proof. By definition of β_{reality} , a reality address requires at least one physical anchor to ground the virtual node. If all parameters t_v , w_v , and p_v are null, the function evaluates to the refusal state $\perp_{\text{reality}}$. This prevents unanchored states from achieving standing. $\square$

Theorem 8.5 (Duality of Virtual-Physical Mapping). *Let LSIF/LSP be the mapping $f_{\text{LSIF}} : \mathcal{S}_{\text{code}} \rightarrow \mathcal{C}_{\text{file}}$ from virtual code symbols to physical file offsets. Let Reality Addressing be the mapping $f_{\text{reality}} : \mathcal{S}_{\text{exec}} \rightarrow \mathcal{C}_{\text{universe}}$ from virtual execution nodes to physical coordinates. Both mappings are structural retractions that preserve topological dependency relations:*

$$v_1 \prec_{\mathcal{E}} v_2 \implies f_{\text{reality}}(v_1) \leq_{\text{time}} f_{\text{reality}}(v_2).$$

Proof. If $v_1 \prec_{\mathcal{E}} v_2$, there exists a causal path in the execution graph, meaning the state transition of v_2 depends on the completed execution of v_1 . Because physical time is monotonic, the physical event corresponding to v_2 cannot occur before v_1 . The public ontology anchor $t_v = f_{\text{reality}}(v).\text{time_anchor}$ must reflect this physical constraint: $t_{v_1} \leq_{\text{time}} t_{v_2}$. Thus, the reality mapping is a homomorphism preserving the causal poset structure on the physical time coordinate. This is dual to LSIF, where compiler dependency order is preserved on physical file layout constraints. $\square$

Chapter 9

The Antibody Clause, Formalized

A thesis addressed to a trillion agents, making claims of planetary capacity, must carry its own guardrail against grandiosity — otherwise it is the very overclaim it warns against. The keystone stated this as the antibody clause. We formalize it as an invariant on claim magnitude and then apply it to this paper.

9.1 Claims, witnesses, and magnitude

Definition 9.1 (Claim, receipt witness, magnitude). A *claim* is a pair $c = (\phi_c, w_c)$ where ϕ_c is a proposition and $w_c \in \mathbb{R}_{\geq 0}$ its asserted *magnitude*. A *receipt witness* for c is a receipt $r(c)$ whose boundary projection is accepted by the fixed verifier, $V(\pi(r(c))) = 1$, and whose committed fields *entail* an attested magnitude $\hat{w}_c$ — the largest value the receipt’s law-bearing fields support under Faithful Projection. A claim is *admissible* iff it has a receipt witness with $w_c \leq \hat{w}_c$.

9.2 Overclaim set and the antibody invariant

Definition 9.2 (Overclaim set and the antibody invariant). The *overclaim set* of a system emitting claims $\mathcal{C}$ is:

$$\mathcal{V} = \{ c \in \mathcal{C} : \nexists r(c) \text{ with } V(\pi(r(c))) = 1 \text{ and } w_c \leq \hat{w}_c \},$$

the claims with no receipt witness, or whose magnitude exceeds what their witness attests. The *antibody invariant* is $\mathcal{V} = \emptyset$: this is the **docs-exceed-mechanism** defect class of the **praxis** definition-of-done, enforced as a gate rather than a guideline.

Theorem 9.1 (Magnitude is bounded by the receipting mechanism). *If a system maintains the antibody invariant $\mathcal{V} = \emptyset$, then for every emitted claim c , its magnitude is bounded by its receipt witness:*

$$w_c \leq \hat{w}_c = \max\{ w : \text{the committed fields of } r(c) \text{ entail magnitude } w \},$$

and $\hat{w}_c$ is verifiable at boundary cost $O(\kappa)$ independent of the interior that produced c . Consequently no claim can exceed what its mechanism can receipt, regardless of its rhetoric.

Proof. By $\mathcal{V} = \emptyset$ (Definition 9.2), every emitted c has a receipt witness with $V(\pi(r(c))) = 1$ and $w_c \leq \hat{w}_c$. The first conjunct is verifiable at $O(\kappa)$ by the Comprehension–Verification Gap and Proposition 2.2 here, and $\hat{w}_c$ is a function of the committed fields, which are faithful under collision resistance. A claim with $w_c > \hat{w}_c$ would lie in $\mathcal{V}$, contradicting the invariant; hence $w_c \leq \hat{w}_c$. The bound is on the magnitude a *receipt* attests, so it is independent of the unread interior. $\square$

9.3 Reflexive antibody: this paper’s own claims

Corollary 9.2 (Reflexive antibody: this paper’s own claims). *Each quantitative claim in this paper is either a theorem with a proof (a receipt in the logical sense: the derivation is its witness, verifiable by reading the proof, of bounded length) or an Estimate explicitly labelled and provisioned with its inputs (a receipt of the second kind: the magnitude is bounded by the stated Fermi inputs, and no larger value is asserted). The partition is exhaustive by construction: no numerical assertion appears unlabelled. Therefore this paper’s overclaim set is empty, and by Theorem 9.1 the magnitude of each claim is bounded by its witness — a proof or a labelled estimate — as the invariant requires.*

Proof. Enumerate the paper’s magnitude-bearing statements. The gap ratio 6.25×10^5 (Theorem 2.1), the footprint 10 GB (Proposition 5.4), the per-word gating 7/64 (Theorem 5.1), and the magnitude bound itself (Theorem 9.1) are theorems with proofs. The wall-time (Estimate 2.1), sweep time (Estimate 5.1), demand, supplies, and affordability separation (Estimates 6.1–6.1) are labelled estimates carrying their inputs. Every magnitude falls in one class or the other; none is unlabelled; hence $\mathcal{V} = \emptyset$ for this document. $\square$

Remark. This is the doctrine’s discipline made self-applying: *a grand system without receipts is grandiosity; a grand system that receipts its own overclaims is machinery.* A paper claiming planetary capacity earns the claim only by receipting it — by proving what it can and labelling what it estimates. The antibody clause is not modesty; it is the mechanism that makes a large claim admissible in the sense of the Chatman equation.

Chapter 10

Fleet Application and Autonomic Reconciliation

10.1 The praxis-retrofit fleet applier

When a plan produces a code modification as its artifact, the artifact must be applied to a live fleet. The `praxis-retrofit` crate realizes this as a *receipted applier*: every application is gated, isolated in a Git worktree, and admitted through a CI oracle before any branch is advanced.

Definition 10.1 (AST-gate isolation). An *AST gate* is a syntactic predicate $g : \mathcal{T} \rightarrow \{0, 1\}$ on an abstract syntax tree $\mathcal{T}$ that is computable in $O(|\mathcal{T}|)$ time. The retrofit applier gates every proposed change through an ordered battery $(g_1, \dots, g_m)$ of AST gates: the change is admitted iff all gates pass, i.e. $\bigwedge_{i=1}^m g_i(\mathcal{T}) = 1$. A failing gate returns a denial with the gate index i and the offending AST node, constituting a *refusal with reason* in the sense of the denial monoid (Proposition 1.2).

Proposition 10.1 (AST-gate admission is total and terminating). *For any finite AST $\mathcal{T}$ and finite gate battery $(g_1, \dots, g_m)$ where each g_i is a syntactic predicate, the battery evaluation terminates in time $O(m \cdot |\mathcal{T}|)$ and returns either **Admitted** or a refusal carrying the first failing gate index and the offending node. The retraction is total: it never diverges and never returns an unlabelled outcome.*

Proof. Each g_i traverses the finite tree $\mathcal{T}$ once in $O(|\mathcal{T}|)$ time. The battery evaluates the gates in order; on the first failure it returns the structured denial; on all-pass it returns **Admitted**. The `RetrofitApplier` in `praxis-retrofit` wraps each gate result in the `DenialPolarity` framework, so the denial composes as a monoid element. $\square$

Definition 10.2 (Isolated worktree applier). The retrofit applier isolates each application attempt in a fresh Git worktree:

`git worktree add temp_path phase_branch,` (10.1)

applies the change, runs the CI oracle $(g_1, \dots, g_m)$ (which includes `cargo build`, `cargo test`, `cargo clippy`), and on any gate failure executes the rollback protocol:

`git reset --hard baseline_sha.`

On all-pass, the worktree change is merged into the target branch, and the chain receipt for the application is minted with the worktree OID committed into the frame's `obj_refs` payload digest.

Proposition 10.2 (Main branch cleanliness is an invariant). *Under Definition 10.2, the main branch is never advanced past a failing CI gate. The rollback protocol is unconditional on failure: it runs before the worktree is removed, restoring the branch to `baseline_sha` regardless of which gate failed. Hence the main branch satisfies all CI gates at every point in its history.*

Proof. The applier's critical section is: (1) create worktree, (2) apply change to worktree, (3) run gate battery, (4) on pass: merge; on fail: `git reset --hard baseline_sha` and abort. Step 4's fail branch executes before the worktree is deleted, and `git reset --hard` on a worktree has no effect on the main working tree. The main branch pointer is advanced only in step 4's pass branch, which is never reached on a gate failure. Hence the main branch history is a subsequence of attempts all of which passed every gate. $\square$

10.2 Visual gap measurement and the reconciler residual

Definition 10.3 (Visual gap report). A *visual gap report* is the output of `praxis-reconciler's` `measure_gap` function: for each reconcilable dimension $i \in \{1, \dots, k\}$, the residual $r_i = \text{measured}_i - \text{midpoint}(\text{target}_i)$ and the dominant dimension $i^* = \arg \max_i |r_i|$ together with a rendered diff block for the discrepancy. The report is bounded: it has k residual values and one dominant index, of total size $O(k)$ independent of the interior that generated the measurements.

Proposition 10.3 (The visual gap report is a bounded projection). *Definition 10.3 gives $\text{gap} : \mathbb{R}^k \rightarrow \mathbb{R}^k \times [k]$, a projection from the k -dimensional measurement space onto the residual vector and its argmax. This projection has the same cost structure as the receipt projection: $O(k)$ to compute, $O(k)$ to verify. The repair operator `apply_repair` then acts on the dominant dimension i^* subject to `RepairBand` bounds, producing at most one corrective actuation — a bounded deterministic manufacture.*

Proof. The residual vector is a linear map (subtraction of midpoint from measurement); it costs $O(k)$ arithmetic operations. The argmax is a single pass over k values. The dominant dimension is one index; the repair acts on that dimension only. By Definition, the `RepairBand` is a finite closed interval, so the repair is a bounded projection onto that interval. $\square$

Chapter 11

Conclusion

The keystone proved that a bounded verifier can trust an unbounded interior through a faithful projection. This paper carried that principle to the scale at which it is not an optimization but a precondition of coherence, and made it quantitative and mechanical at each step.

We measured the gap: on the canonical 2.5-million-token instance it is $\sim 6 \times 10^5$, with measured inputs and a proved consequence (Theorem 2.1), and boundary cost is provably invariant as the interior grows (Proposition 2.2). We formalized manufactured trust as the decidable-oracle move (Proposition 3.1) and gave it three faces — differential agreement with a false-accept bound (Proposition 3.2), boundary fuzzing of a total retraction (Proposition 3.3), and mutation-testing with the Mutant Kill Theorem (Theorem 3.4) that ties a kill to rejection at the correct stage and turns the verifier’s stage index into a diagnosis (Corollary 3.5). We built the agent byte from the running denial lanes (Construction 5.1) and proved fleet admission is branchless mask algebra gating 64 agents per word in seven instructions (Theorem 5.1), with a 10^{10} -agent population an exact 10 GB swept in one bandwidth-bound pass (Proposition 5.4, Estimate 5.1). We weighed the planetary economics and proved the affordability separation (Theorem 6.1): comprehension-based admission is under-provisioned by three to four orders of magnitude, and only bit-parallel admission scales (Corollary 6.2). Finally we formalized the antibody clause as the invariant $\mathcal{V} = \emptyset$ and proved that claim magnitude is bounded by the receipting mechanism (Theorem 9.1), applying the bound reflexively to this paper (Corollary 9.2).

The picture that results is severe and simple. At planetary scale, no one reads anyone. Trust is not comprehension; it is a hash comparison, an oracle equality, a total classifier’s label, and — for the population as a whole — a single AND of eight bit-planes streamed across memory. The interior may be a trillion tokens or a trillion agents; the admission is a sweep, and the sweep is affordable. What makes the civilization coherent is not that its members understand one another. It is that each carries a faithful byte, and the byte can be checked in one pass.

Beyond cognitive load; within admissible projection — at the scale of the sweep.

Appendix A

Notation and Constants

Symbols specific to this paper

Symbol	Meaning
$T(\sigma)$	interior token count of a manufacture (measured)
κ	boundary field count / verifier context capacity (design constant)
t_H	per-frame recomputation wall-time (estimated)
$\Omega, \Omega_{f,g}, \Omega_{\partial}$	decidable oracle; differential; admission-boundary
$V = (V_1, \dots, V_k)$	staged validator; I_i its stage invariants
$s(m), \text{Kill}(m)$	correct stage of a mutation; kill indicator
$b_a \in \{0, 1\}^8$	agent status byte (Construction 5.1)
P_ℓ, D_ℓ	OK-polarity and denial-polarity bit-planes of lane ℓ
B	streaming memory bandwidth
$\Lambda_{\text{demand}}, \Lambda_{\text{comp}}, \Lambda_{\text{bit}}$	admission demand; comprehension supply; bit-parallel supply
$\mathcal{V}, w_c, \hat{w}_c$	overclaim set; claimed and attested magnitude

Constants used, with provenance

Quantity	Value	Provenance
Interior tokens $T(\sigma)$	$\approx 2.5 \times 10^6$	measured (keystone Ch. 6)
Boundary fields κ	4	design (receipt schema)
Gap ratio T/κ	$\approx 6.25 \times 10^5$	proved (Thm. 2.1)
Byte width	8 bits	design (denial lanes)
Gating cost	7/64 instr/agent	proved (Thm. 5.1)
10^{10} -agent footprint	10 GB	exact (Prop. 5.4)
Sweep time	25–100 ms	estimate (Est. 5.1)
Bit-parallel supply	$\sim 10^{11}$ /s/server	estimate (Cor. 5.5)
Comprehension supply	$\sim 10^8$ /s planetary	estimate (Est. 6.2)
Per-decision cost ratio	$\sim 10^7$	keystone (Prop. 6.2)

Bibliography

- [1] The Praxis / Capability Physics Program, *The Projection Principle: A First-Principles Theory of Bounded Receipted Execution for a Trillion Agents Beyond One Another's Comprehension* (keystone thesis of this series), 2026.
- [2] H. G. Rice, *Classes of recursively enumerable sets and their decision problems*, Transactions of the American Mathematical Society **74** (1953), 358–366.
- [3] G. Langdale and D. Lemire, *Parsing gigabytes of JSON per second*, The VLDB Journal **28**(6) (2019), 941–960.
- [4] H. Bast and B. Buchhold, *QLever: A query engine for efficient SPARQL+Text search*, in Proc. CIKM, ACM, 2017, 647–656.
- [5] R. A. DeMillo, R. J. Lipton, and F. G. Sayward, *Hints on test data selection: Help for the practicing programmer*, Computer **11**(4) (1978), 34–41.
- [6] Y. Jia and M. Harman, *An analysis and survey of the development of mutation testing*, IEEE Transactions on Software Engineering **37**(5) (2011), 649–678.
- [7] J. O'Connor, J.-P. Aumasson, S. Neves, and Z. Wilcox-O'Hearn, *BLAKE3: One function, fast everywhere*, 2020.
- [8] C. E. Shannon, *A Mathematical Theory of Communication*, Bell System Technical Journal extbf27 (1948), 379–423 & 623–656.
- [9] S. Banach, *Sur les fonctionnelles linéaires*, Studia Mathematica extbf1 (1929), 211–216.
- [10] S. Mac Lane, *Natural associativity and commutativity*, Rice University Studies extbf49(4) (1963), 28–46.
- [11] G. A. Miller, *The magical number seven, plus or minus two: Some limits on our capacity for processing information*, Psychological Review extbf63(2) (1956), 81–97.
- [12] N. Cowan, *The magical number 4 in short-term memory: A reconsideration of mental storage capacity*, Behavioral and Brain Sciences extbf24(1) (2001), 87–114.
- [13] L. Lamport, *Time, clocks, and the ordering of events in a distributed system*, Communications of the ACM extbf21(7) (1978), 558–565.
- [14] C. J. Fidge, *Timestamps in Message-Passing Systems That Preserve the Partial Ordering*, in Proc. 11th Australian Computer Science Conference extbf10(1) (1988), 56–66.

- [15] F. Mattern, *Virtual Time and Global States of Distributed Systems*, Parallel and Distributed Algorithms (1989), 215–226.
- [16] J.-M. Lasry and P.-L. Lions, *Mean field games*, Japanese Journal of Mathematics extbf2(1) (2007), 229–260.
- [17] M. Huang, R. P. Malham’e, and P. E. Caines, *Large population stochastic dynamic games: closed-loop McKean-Vlasov systems and the Nash certainty equivalence principle*, Communications in Information and Systems extbf6(3) (2006), 221–252.
- [18] A. Fick, *Ueber Diffusion*, Annalen der Physik extbf170(1) (1855), 59–86.
- [19] C.-L. Navier, *M’emoire sur les lois du mouvement des fluides, en ayant ’egard à l’adh’esion des mol’ecules*, M’emoires de l’Acad’emie Royale des Sciences de l’Institut de France extbf6 (1822), 389–440.
- [20] G. G. Stokes, *On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of elastic solids*, Transactions of the Cambridge Philosophical Society extbf8 (1845), 287–305.
- [21] R. Bellman, *Dynamic Programming*, Princeton University Press, 1957.
- [22] H. Nyquist, *Regeneration Theory*, Bell System Technical Journal extbf11(1) (1932), 126–147.
- [23] Microsoft Corporation, *Language Server Protocol Specification*, GitHub Open Specification, 2016.
- [24] Microsoft Corporation, *Language Server Index Format Specification*, LSP Specifications, 2019.
- [25] Sourcegraph, Inc., *SCIP: Source Code Intelligence Protocol*, GitHub Specification, 2022.
- [26] W3C, *Time Ontology in OWL*, W3C Recommendation, 2006.
- [27] OGC, *OGC GeoSPARQL - A Geographic Query Language for RDF Data*, Open Geospatial Consortium, 2012.
- [28] W3C, *PROV-O: The PROV Ontology*, W3C Recommendation, 2013.
- [29] P. Todd, *Merkle Mountain Ranges*, GitHub Specification, 2012.
- [30] R. Merkle, *Secrecy, authentication, and public key systems*, Stanford University Thesis, 1979.
- [31] J. Benaloh and M. de Mare, *One-way accumulators: A decentralized alternative to digital signatures*, in Proc. EUROCRYPT (1993), 274–285.